PROFESSIONAL INSTALLATION,
OPERATING AND SERVICE MANUAL

NITRATE CONDITIONER

- ☐ BP150NJ
- ☐ BP150NJBF

Certified by IAPMO R&T to:
NSF/ANSI STANDARD 53
ASSE LEC 2008
NSF/ANSI/CAN 372



PLATINUM SERIES

ELECTRONIC NITRATE CONDITIONER
PROGRAMMED & EQUIPPED WITH A
"NO NITRATE WATER BYPASS VALVE"
TO PREVENT UNTREATED WATER BYPASS
DURING REGENERATION

LEAD FREE
UPC

IAPMO R&T
WATER QUALITY
PLATINUM SEAL
IAPMO RESEARCH AND TESTING

Antimicrobial Technology

BRITA|PRO

JOB SPECIFICATIONS

MODEL NO. _____

INSTALLATION DATE _____

SERIAL NUMBER _____

INSTALLER NAME _____ PHONE _____

ADDRESS _____

WATER TEST AT TIME OF INSTALLATION

_____ Incoming PSI

_____ Hardness $CaCo_3$ (gpg)

_____ Iron (ppm)

_____ pH

Nitrate NO_3-N (mg/L): _____

Other: _____

_____

Notes -

gpg: grains per gallon

ppm: parts per million

mg/L: milligrams per liter

1ppm = 1 mg/L

MISC. INFORMATION

All Water is Treated Except:

_____ Rear Hose Bib _____ Front Hose Bib _____ Toilets

_____ Other _____

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GENERAL INSTALLATION AND SERVICE WARNINGS

The control valve, fittings and/or bypass are designed to accommodate minor plumbing misalignments but are not designed to support the weight of a system or the plumbing.

HYDROCARBONS SUCH AS KEROSENE, BENZENE, GASOLINE, ETC., MAY DAMAGE PRODUCTS THAT CONTAIN O-RINGS OR PLASTIC COMPONENTS. EXPOSURE TO SUCH HYDROCARBONS MAY CAUSE THE PRODUCTS TO LEAK. DO NOT USE THE PRODUCT(S) CONTAINED IN THIS DOCUMENT ON WATER SUPPLIES THAT CONTAIN HYDROCARBONS SUCH AS KEROSENE, BENZENE, GASOLINE, ETC.

Do not use Vaseline, oils, other hydrocarbon lubricants or spray silicone anywhere. A silicone lubricant may be used on black o-rings.

The nuts and caps are designed to be unscrewed or tightened by hand only or with the clack service wrench. Do not use a pipe wrench to tighten or loosen nuts or caps. Do not place a screwdriver in the slots on caps and/or tap with a hammer.
Do not use pipe dope or other sealants on threads. Use Teflon tape on the threaded inlet, outlet and drain fittings. Teflon tape is not necessary on the nut connection or caps because of o-ring seals.

After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons, unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3+ seconds. This resets the electronics and establishes the service piston position. The display should flash all wording, then flash the software version and then reset the valve to the service position.

All plumbing should be done in accordance with local plumbing codes. The pipe size for the drain line should be a minimum of ½" (ID). Backwash flow rates in excess of 7 gpm or length in excess of 20' require ¾" drain line.

Solder joints near the drain must be done prior to connecting the drain line flow control fitting. Leave at least 6" between the drain line control fitting and solder joints when soldering pipes that are connected on the drain line control fitting. Failure to do this could cause interior damage to the drain line flow control fitting.

When assembling the installation fitting package (inlet and outlet), connect the fitting to the plumbing system first and then attach the nut, split ring and o-ring. Heat from soldering or solvent cements may damage the nut, split ring or o-ring. Solder joints should be cool and solvent cements should be set before installing the nut, split ring and o-ring. Avoid getting primer and solvent cement on any part of the o-rings, split rings, bypass valve or control valve.

Plug into an electrical outlet. Note: All electrical connections must be connected according to local codes. (Be certain the outlet is uninterrupted.)

Install grounding strap on metal pipes.

This glass filled Noryl (or equivalent) fully automatic control valve is designed as the primary control center to direct and regulate all cycles of a water nitrate treatment system. The control valve intentionally does not have a built-in water meter and is programmed to regenerate as a time clock (passage of a particular number of days). **THE WATER METER HAS BEEN REMOVED BY THE FACTORY BECAUSE IT SHOULD NOT BE USED AS THE PRIMARY MONITORING DEVICE FOR CRITICAL OR HEALTH EFFECT APPLICATIONS.**

It is not recommended to change control valves from downflow to upflow brining or vice versa in the field. The valve bodies for downflow and upflow are unique to the regeneration type and should not be interchanged. A mismatch of valve body and regeneration piston will result in untreated water bypass during service.

The control valve is compatible with a variety of regenerants and resin cleaners. The control valve is capable of routing the flow of water in the necessary paths to regenerate or backwash water treatment systems. The injector regulates the flow of brine or other regenerants. The control valve regulates the flow rates for backwashing, rinsing, and the replenishing of treated water into a regenerant tank, when applicable. The control valve uses no traditional fasteners (e.g. screws); instead clips, threaded caps and nuts and snap type latches are used. Caps and nuts only need to be firmly hand tightened because radial seals are used. Tools required to service equipment is the clack service wrench. Disassembly for servicing takes much less time than comparable products currently on the market. Control valve installation is made easy because the distributor tube can be cut ½" above to ½" below the top of tank thread. The distributor tube is held in place by an o-ring seal and the control valve also has a bayonet lock feature for upper distributor baskets.

The AC adapter comes with a 15 foot power cord and is designed for use with the control valve. The AC adapter is for dry location use only. The control valve remembers all settings until the battery power is depleted if the power goes out. After the battery power is depleted, the only item that needs to be reset is the time of day; other values are permanently stored in the nonvolatile memory. The control valve battery is not rechargeable but is replaceable.

Allow one foot of clearance to service the valve.

If this device is not maintained and operated as specified in the owner's manual, there is a risk of exposure to contaminants.

PRE-INSTALLATION REVIEW

WATER QUALITY: If sand or sediment is present in the water supply, a sediment filter should be installed ahead of the nitrate treatment system. Air must not come in contact with water until after it has passed through the nitrate treatment system. In some cases, additional treatment equipment may be needed to treat water having special characteristics, such as: ferric hydroxide iron, iron bacteria, low pH, tastes and odors, hardness, etc. Consult your dealer if you have any questions. **This nitrate treatment system is not used for treating water that is microbiologically unsafe or of unknown quality without adequate disinfection before or after treatment.**

PRE-TREATMENT
Filtration may be required to remove Iron, Manganese, turbidity, total suspended solids (TSS) and organic matter as these could foul the resin.
Water Softening will be required for hardness levels of 4 grains or higher to prevent scaling.
Dechlorination of Chlorine and Chloramines will be required to prevent oxidation degradation of the resin.

POST-TREATMENT
pH adjustment may be required after the unit to avoid corrosion. Nitrate and other anions displace chlorine on the resin, releasing the chloride to the product water. Chloride can cause taste issues, dezincification, and galvanic corrosion.

IMPORTANT:

NITRATES MUST BE TESTED BEFORE INSTALLATION AS MILLIGRAMS PER LITER (MG/L) AS NITRATE-NITROGEN.

REGULARLY MONITOR NITRATE LEVELS BEFORE AND AFTER SYSTEM.

NITRATE LEVELS CAN CHANGE OVER TIME.

ON PAGE 14 OF THIS MANUAL, A TABLE IS PROVIDED FOR THE BP150NJ MODEL. TO USE THIS TABLE, USE THE LATEST NITRATE CONCENTRATIONS, AND IN CONJUNCTION WITH THE NUMBER OF PEOPLE USING THE NITRATE TREATMENT SYSTEM, DETERMINE THE NUMBER OF DAYS (DAYS OVERRIDE) BETWEEN THE REGENERATION OF THE NR-1 RESIN IN THE TREATMENT SYSTEM.

USE A HOME TEST KIT TO MONITOR LEVELS MONTHLY AS WELL AS HAVING YOUR WATER PROFESSIONALLY TESTED FOR NITRATE QUARTERLY OR AT LEAST ANNUALLY.

EFFICIENCY OF NITRATE REMOVAL NORMALLY DEPENDS ON THE PRESENCE OF COMPETING ANIONS, ESPECIALLY SULFATES, WHICH CAN CAUSE THE DUMPING OF THE NITRATES. HOWEVER RESINEX ™ NR-1 ANION EXCHANGE RESIN USED IN THE BP150NJ TREATMENT SYSTEM, HAS BEEN SPECIALLY DEVELOPED FOR SELECTIVE NITRATE REMOVAL FROM POTABLE WATER IN THE PRESENCE OF HIGH LEVELS OF SULFATE.

THUS THERE IS NO DANGER OF NITRATE DUMPING IN THIS NITRATE REMOVAL SYSTEM.

WATER PRESSURE: A minimum of 30 pounds of water pressure (psi) is required for regeneration. Maximum 80 psi. CAUTION: the nitrate treatment system cannot be subject to a vacuum due to loss of pressure (such as a water main break or submersible well pump check valve failure).

WATER TEMPERATURE: The range of water temperature is 40º F to 100º F. **DO NOT** install any nitrate conditioner with less than 10 feet of piping between its outlet and the inlet of a water heater.

AMBIENT TEMPERATURE: DO NOT locate nitrate treatment system where it or its connections (including the drain and overflow lines) will ever be subject to room temperatures under 33º F.

ELECTRICITY: An uninterrupted 120 volt 60Hz source is required. *Make sure electrical source is not on a timer or switch.* All electrical connections must be connected according to local codes. The plug-in transformer is for dry locations only. Surge protection is recomended with all electrical connections.

DRAIN: All plumbing should be done in accordance with local plumbing codes. The distance between the drain and the nitrate treatment system should be as short as possible. The pipe size for the drain line should be a minimum of 1/2 " (inside diameter of pipe).

NITRATE REDUCTION: It is recommended that the nitrate treatment system be installed to treat both the hot and cold water supply. Outside faucets for non-potable use can be left on untreated water.

BYPASS: A bypass valve should be installed so that water will be available if it should be necessary to shut off the pressure in order to service the nitrate treatment system.

NO NITRATE WATER BYPASS VALVE: The nitrate treatment system is programmed and equipped with a "No Nitrate Water Bypass Valve" to prevent untreated water bypass during regeneration.

BYPASS VALVE

The bypass valve is typically used to isolate the control valve from the plumbing system's water pressure in order to perform control valve repairs or maintenance. The bypass valve is particularly unique in the water treatment industry due to its versatility and state of the art design features. The 1" full flow bypass valve incorporates four positions, including a diagnostic position that allows service personnel to work on a pressurized system while still providing untreated bypassed water to the facility or residence. Its completely non-metallic, all-plastic design allows for easy access and serviceability without the need for tools.

The bypass body and rotors are glass filled Noryl® (or equivalent) and the nuts and caps are glass filled polypropylene. All seals are self-lubricating EPDM to help prevent valve seizing after long periods of non-use. Internal o-rings can easily be replaced if service is required.

The bypass consists of two interchangeable plug valves that are operated independently by red arrow-shaped handles. The handles identify the flow direction of the water. The plug valves enable the bypass valve to operate in four positions.

OPERATION:

1. Service Operation Position: The inlet and outlet handles point in the direction of flow indicated by the engraved arrows on the control valve. Water flows through the control valve during normal operation and this position also allows the control valve to isolate the media bed during the regeneration cycle. *(see figure 1)*

2. Bypass Position: The inlet and outlet handles point to the center of the bypass, the control valve is isolated from the water pressure contained in the plumbing system. Untreated water is supplied to the plumbing system. *(see figure 2)*

3. Diagnostic Position: The inlet handle points in the direction of flow and the outlet handle points to the center of bypass valve, system water pressure is allowed to the control valve and the plumbing system while not allowing water to exit from the control valve to the plumbing. *(see figure 3)*

4. Shut Off Position: The inlet handle points to the center of the bypass valve and the outlet handle points in the direction of flow, the water is shut off to the plumbing system. If water is available on the outlet side of the nitrate conditioner it is an indication of water bypass around the system (i.e. a plumbing connection somewhere in the building bypasses the system). *(see figure 4)*

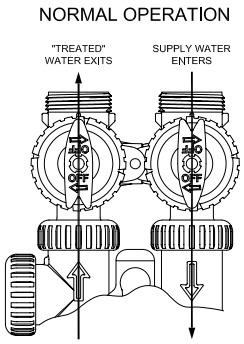


figure 1

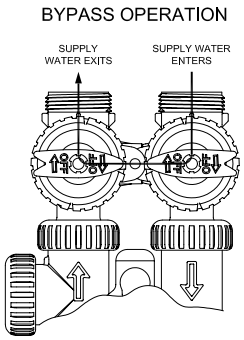


figure 2

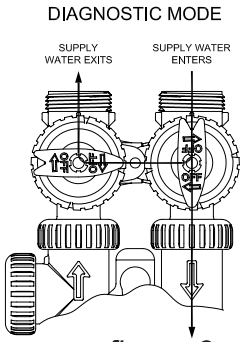


figure 3

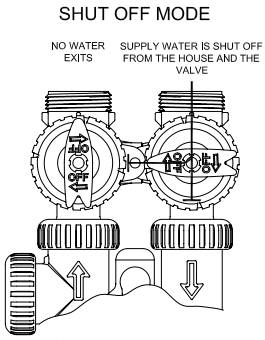
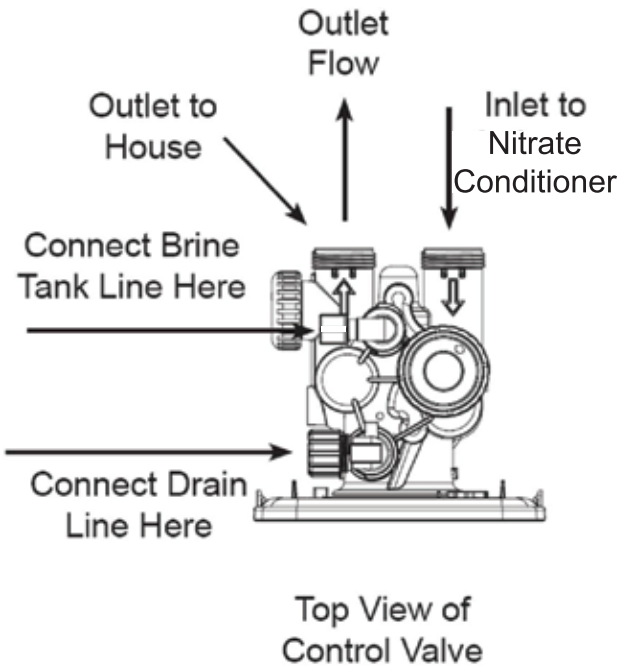


figure 4

The nuts and caps are designed to be unscrewed or tightened by hand or the use of a clack service wrench is recommended. Do not use a pipe wrench to tighten or loosen nuts or caps. Do not place screwdriver in slots on caps and/or tap with a hammer. To access the rotor, unscrew the cap and lift the cap, rotor and handle out as one unit. Twisting the unit as you pull it out will help to remove it more easily. There are three o-rings: one under the rotor cap, one on the rotor stem and the rotor seal. Replace worn o-rings. Clean rotor. Reinstall rotor.

INSTALLATION INSTRUCTIONS

(All electrical & plumbing should be done in accordance to all local codes)

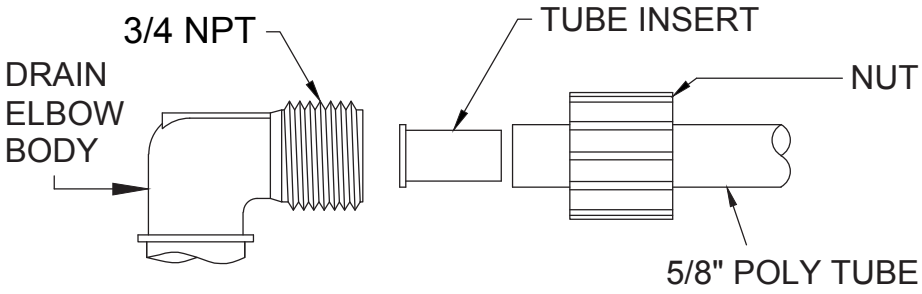


1. Place the nitrate treatment system where you want to install it, making sure it is on a clean, level and firm base.

2. Do all necessary plumbing (inlet to inlet, outlet to outlet, and drain line to drain). The control valve, fittings and/or bypass are designed to accommodate minor plumbing misalignments but are not designed to support the weight of a system or the plumbing.

3. When assembling the installation fitting package (inlet and outlet), connect the fitting to the plumbing system first and then attach the nut, split ring and o-ring. Heat from soldering or solvent cements may damage the nut, split ring or o-ring. Solder joint should be cool and solvent cements should be set before installing the nut, split ring and o-ring. Avoid getting primer and solvent cement on any part of the o-rings, split rings, bypass valve or control valve.

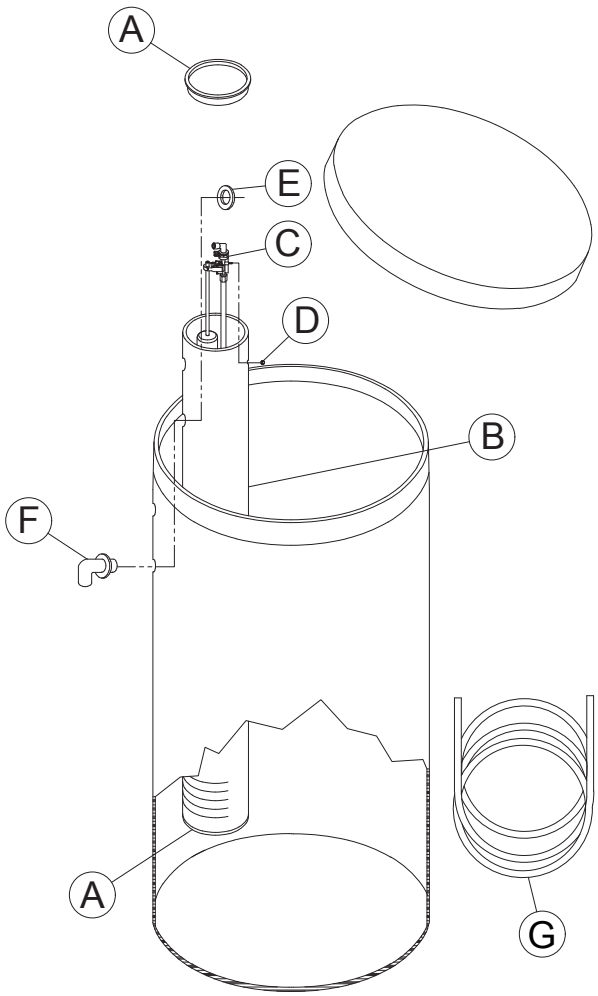
4. A jumper ground wire should be installed between the inlet and outlet pipe whenever the metallic continuity of a water distribution piping system is interrupted. Install grounding strap on metal pipes.



DRAIN LINE FITTING CONNECTION

5. The drain connection may be made using either 5/8” polytube (see above) or a 3/4” female adapter. The polytube insert is shipped attached to the drain line elbow’s locking clip. Press the insert into the drain line tubing (tubing not provided). Loosen the nut of the drain line elbow. Press the 5/8” polytube with insert into the drain line elbow until it seats on the back of the fitting. Tighten the nut. If soldering, joints near the drain must be done prior to connecting the drain line flow control fitting. Leave at least 6” between the drain line control fitting and solder joints when soldering pipes that are connected on the drain line control fitting. Failure to do this could cause interior damage to the drain line flow control fitting. Never insert a drain line into a drain, sewer line, or trap. Always allow an air gap between the drain line and the wastewater to prevent the possibility of sewage being back-siphoned into the nitrate conditioner.

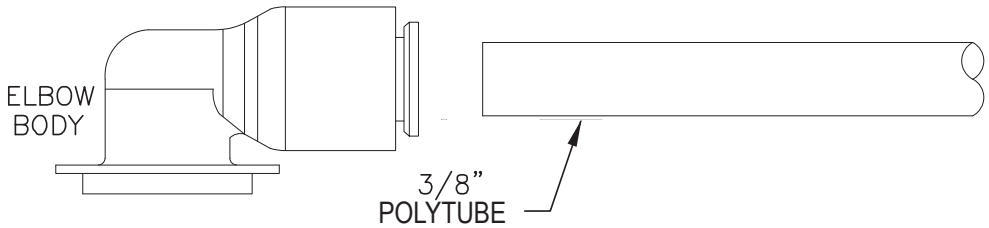
(All electrical & plumbing should be done in accordance to all local codes)



6. Assemble the brine tank. Install a brine well cap (Item A) on the bottom of the brine well (Item B). Remove the brine valve assembly (Item C) from the brine well by removing the nut (Item D). Attach the brine well to the brine tank with the two-piece overflow (Items E & F) using the lower holes in both pieces, as shown. Reinstall the brine valve assembly. Push brine line polytube (Item G) through the upper holes in the brine tank and brine well and press the polytube into the elbow fitting.

7. The brine refill flow control assembly is installed in an easy to access refill elbow located on top of the control valve. The refill flow control assembly is attached to the control valve with a locking clip. The locking clip allows the elbow to rotate 270 degrees so the outlet can be oriented towards the brine tank.

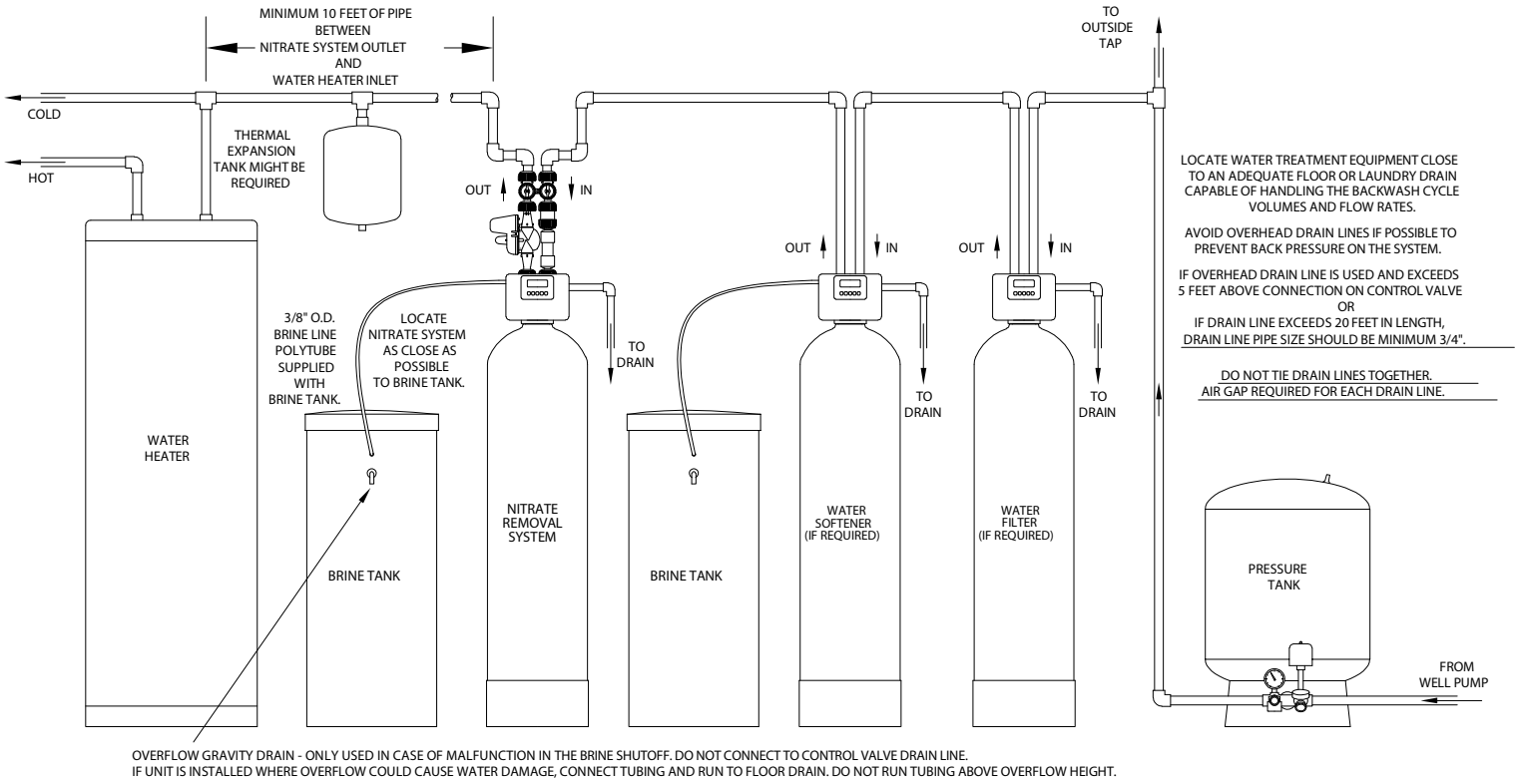
8. Connect the brine line polytubing found with the brine tank to the brine connection on the control valve. The control valve has a standard refill elbow to which a 3/8" flexible tube can be connected, see below. Press the polytube into the brine elbow fitting. Make sure the floor is clean beneath the brine tank and that it is level and smooth.



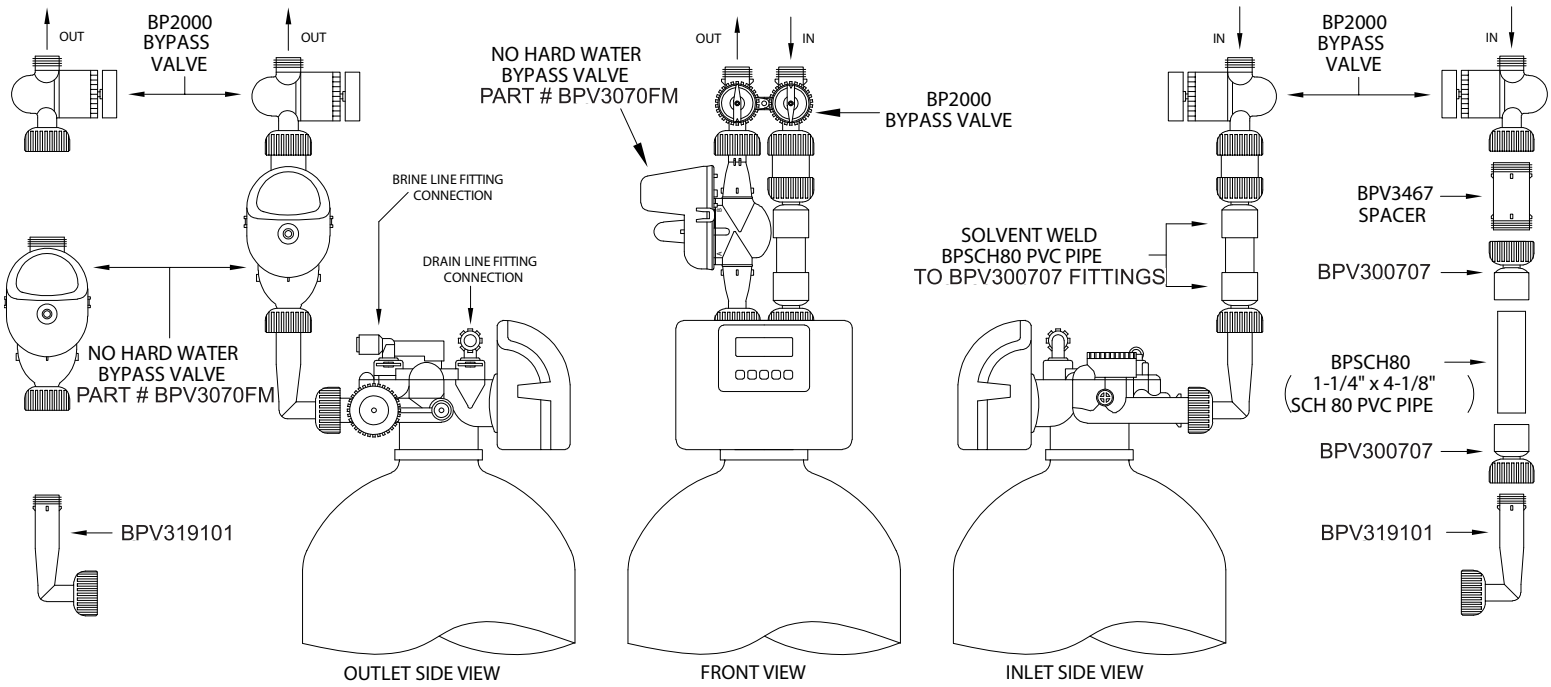
BRINE LINE FITTING CONNECTIONS

9. A 1/2" (inside diameter, not provided) gravity drain line should be connected to the overflow fitting on the side of the brine tank. This overflow is in case of a malfunction in the brine shut off. If the unit is installed where water may flow in the event of an overflow and cause water damage, connect a length of flexible tubing and run to a drain below the level of the overflow. **(Do not connect the tubing to the drain line on the control valve. Do not run tubing above overflow height at any point.)**

BRITA PRO NITRATE REMOVAL SYSTEM INSTALLATION



NITRATE REMOVAL SYSTEM ENLARGED VIEWS



NOTE:
The No Nitrate Water Bypass Valve is open when the threaded portion of its drive gear is made visible in the clear dome of its cover.

CONNECTING TO NO NITRATE WATER BYPASS VALVE

The control valve is factory programmed to have no untreated water bypass during regeneration. The No Nitrate Water Bypass Valve cable MUST be connected to the control valve PC Board BEFORE the control valve transformer is plugged into a 120 volt receptacle. The control valve must recognize the No Nitrate Water Bypass Valve connection, otherwise error code 106 will display.

1. Plumb the No Nitrate Water Bypass Valve according to the drawing on page 9.



2. Remove the front cover from the control valve. Pull out on the cover release tabs located on each side of the cover and pull cover off.



3. Disconnect power and motor cables from PC Board.



4. Remove the drive bracket assembly by pressing up on the drive bracket release tabs and pulling out on the top tabs; the drive bracket should come forward for removal.



5. Bring through the backplate the No Nitrate Water Bypass Valve cable as shown.



6. Approximately 1 ft. of cable should be allowed through the backplate. Weave the cable through the strain relief and above the drive bracket release tabs as shown.



7. Re-install the drive bracket assembly. Route and connect the No Nitrate Water Bypass Valve cable to the two pin connector labeled " MAV" on the PC Board and re-connect motor and power cables as shown. Replace the front cover of control valve.

PLACING UNIT INTO SERVICE

Do not add salt to the brine tank yet. Do not plug the transformer into the receptacle yet. Place bypass in the bypass position. Turn on main water supply. Open a cold water faucet. This will clear the line of any debris (solder, pipe dope, etc.) that may be in the line. Let water run at faucet for a couple minutes, or until clear. Turn off faucet. Manually add 1 ½ gallons of water to the brine tank. Now plug the transformer into a 120 volt receptacle (be certain the receptacle is uninterrupted). Within 5 seconds the control display and buttons will illuminate and the time of day screen will appear.

- Press and hold the REGEN button for approximately 5 seconds until the motor starts.
- The No Nitrate Water Bypass Valve will close.
- Wait until display reads **BACKWASH** and numbers start counting down.
- Momentarily press REGEN again. Valve is now in the **REGENERANT DRAW DN** position.
- Momentarily press REGEN again. Valve is now in the second **BACKWASH** position.
- SLOWLY turn bypass 1/4 turn to **DIAGNOSTIC** mode position (*see figure 3 on page 6*) to allow water to slowly enter Nitrate water treatment system.

When water is flowing steadily to drain without the presence of air, momentarily press REGEN again. Display will read **RINSE** and leave in diamagnetic mode for 30-60 seconds before slowly putting system into service mode (*see figure 1 on page 6*). Allow control to finish the **RINSE** cycle. The No Nitrate Water Bypass Valve will open at the end of the **RINSE** cycle. Allow the control to automatically advance to the **FILL** position. The brine tank will now automatically fill with the proper volume of water for the first regeneration. Allow the control to automatically advance to the **SOFTENING** (nitrate reduction) position. Load the brine tank with salt. Solar Salt is recommended. The brine tank salt level should be checked every couple of weeks to determine salt usage. Keeping the brine tank salt level at least 1/2 full is recommended.

SANITIZING: Use 2 oz. of 5-1/4 % unscented household chlorine bleach for each cubic foot of resin. Pour bleach directly into the 4" diameter white brine well assembly tube located inside the brine tank (*Please refer page 8 labeled B*). Press and hold the REGEN for 5-6 seconds until the motor starts running. Allow system to complete the regeneration automatically. Check for other local and state codes which may also specify sanitation methods.

GENERAL OPERATION

Note: As an energy saving feature, the control will automatically turn off all SOLID BLUE or SOLID GREEN display illumination and keypad illumination after about 5 minutes of the last keypad button push. Any further keypad touch will cause the re-illumination of the display and keypad, and re-activate keypad control.

User Displays

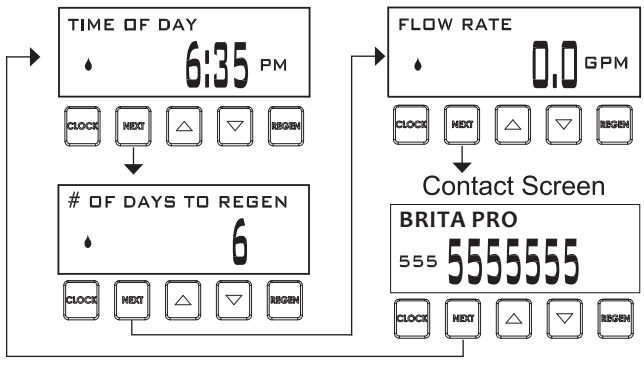
When the system is in normal service mode, one of up to four available User Displays will be shown. Pressing NEXT will alternate between the following displays:

- Current time of day
- Treated water flow rate (see Note below)
- Service contact name and phone number (if entered)
- Remaining days to regeneration

Pressing the ▼ button while in the Days Remaining display will decrease the days remaining in one day increments.

To clear the Service Call reminder, press the ▲ and ▼ buttons simultaneously while the number and banner text screen is displayed.

If the system has called for a regeneration that will occur at the preset time of regeneration, the words REGEN TODAY will alternate with the header on the display.



NOTE: THE CONTROL VALVE'S WATER METER HAS BEEN INTENTIONALLY REMOVED BY THE FACTORY BECAUSE IT SHOULD NOT BE USED AS THE PRIMARY MONITORING DEVICE FOR CRITICAL OR HEALTH EFFECT APPLICATIONS. As a result, the treated water flow rate display will always read 0.0 GPM. The control valve is programmed to regenerate based on the passage of a particular number of days.

SET TIME OF DAY

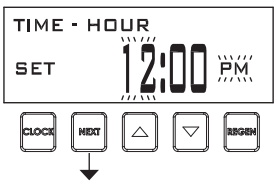
Current time of day needs to be entered during initial installation, and adjusted when daylight saving time begins or ends. If an extended power outage occurs and depletes the on-board non-rechargeable coin cell battery, when power resumes the time of day should be reset and battery replaced.

STEP 1



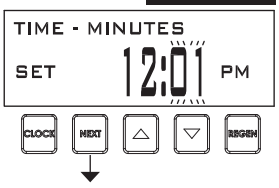
STEP 1 – Press CLOCK.

STEP 2



STEP 2 - Current Time (hour): Set the hour of the day using ▼ or ▲. AM/PM toggles after 12. Press NEXT to go to Step 3.

STEP 3

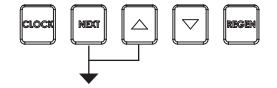


STEP 3 - Current Time (minutes): Set the minutes of the day using ▼ or ▲. Press NEXT to exit Set Time of Day. Press REGEN to return to previous step.

RETURN TO NORMAL MODE

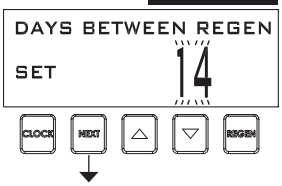
ADJUST DAYS BETWEEN REGENERATIONS OR TIME OF REGENERATION

STEP 1



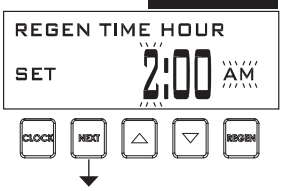
STEP 1 - Press NEXT and ▲ simultaneously for 3+ seconds.

STEP 2



STEP 2 – Day Override: Set the maximum number of days between regenerations (allowable range from 1 to 28). Set Day Override using ▲ or ▼:
Refer to Days Override Selection Chart on page 14
Press NEXT to go to step 3. Press REGEN to return to previous step.

STEP 3



STEP 3 – Next Regeneration Time (hour): Set the hour of day for regeneration using ▲ or ▼ buttons. AM/PM toggles after 12. The default time is 2:00 AM. Press NEXT to go to Step 4. Press REGEN to return to previous step.

STEP 4



STEP 4 – Next Regeneration Time (minutes): Set the minutes of day for regeneration using ▲ or ▼ buttons. Press NEXT exit Installer Display settings. Press REGEN to return to previous step.

RETURN TO NORMAL MODE

Model Number: BP150NJ

Number of People	Number of Days Between Regenerations (Days Override)								Gallons per Day Water Usage
10	16	8	5	4	3	2	2	2	**700**
9	18	9	6	4	3	3	2	2	**630**
8	21	10	7	5	4	3	3	2	**560**
7	24	12	8	6	4	4	3	3	**490**
6	28	14	9	7	5	4	4	3	**420**
5	28	16	11	8	6	5	4	4	**350**
4	28	21	14	10	8	7	6	5	**280**
3	28	28	18	14	11	9	8	7	**210**
2	28	28	28	21	16	14	12	10	**140**
1	28	28	28	28	28	28	24	21	**70**
	5	**10**	**15**	**20**	**25**	**30**	**35**	**40**	
	Nitrate (measured as Nitrogen) NO3-N mg/L								

Important: Regularly monitor nitrate levels before and after system. Nitrate levels can change over time. Use a home test kit to monitor levels monthly as well as having your water professionally tested for nitrate quarterly or at least annually.

Note 1: Nitrates are often measured by laboratories as Nitrogen.

Note 2: This chart is based on 70 gallons per person per day water use.

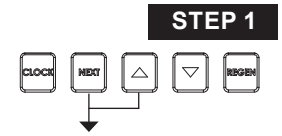
Note 3: This chart represents an approximation of the number of days between regenerations based on 8,241 grains/cu. ft. as CaCO3 operating capacity of Jacobi Resinex NR-1 for nitrate removal at 10 pounds of NaCl per cu. ft. with an influent concentration of 600 mg/L TDS (sulfate concentration of 265 mg/L).

Note 4: Days Override Feature: 1 - 28 days available.

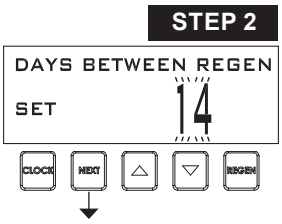
LOW BATTERY

A non-rechargeable coin cell battery is located on the circuit board, used only to maintain the time of day during power outages (all other information will be stored in memory no matter how long the power outage). The screen displays LOW BATTERY when the battery needs to be replaced. The screen will remain illuminated solid blue when LOW BATTERY is displayed. Initially LOW BATTERY display will alternate with the User display, finally displaying only LOW BATTERY. User displays are still accessible by pressing NEXT.

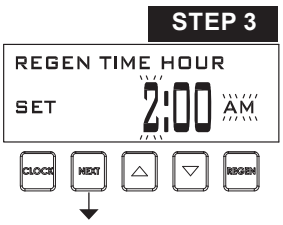
CONTACT SCREEN PROGRAMMING



STEP 1 - Press NEXT and ▲ simultaneously for 3+ seconds to access Installer Display Settings.



STEP 2 – Press NEXT to go to step 3.



STEP 3 – While hour is flashing, press and hold both CLOCK and ▲ to change phone number and banner text.



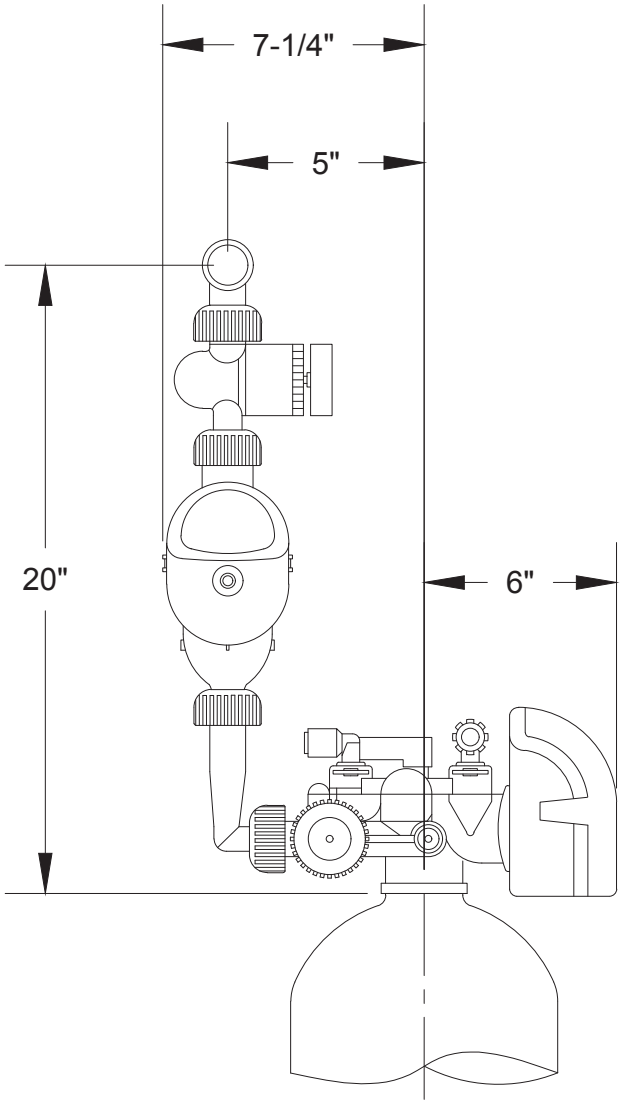
Phone Number - Set phone number using ▲ or ▼. Press NEXT to forward to the next digit. Press REGEN to return to previous digit.



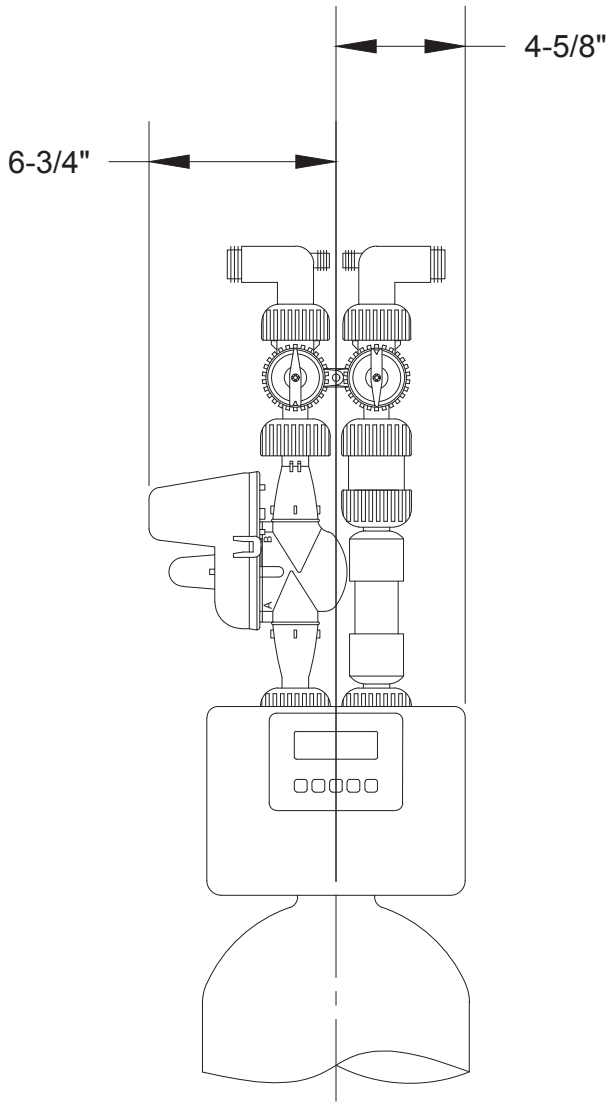
Banner Text - Set the banner text up to a maximum of 44 characters. Use ▲ or ▼ to select letters of the alphabet, number, ampersand (&), or space in the banner text. Press NEXT to forward to the next character or to exit the Installer Display Settings.

GENERAL SPECIFICATIONS

Inlet/Outlet Fitting Options (inches)	3/4 to 1-1/4
Drain Pipe NPT or OD Poly Tube Size (inches)	3/4 NPT or 5/8 OD
Working Pressure; Min to Max (PSI)	30 to 80
Water Operating Temp; Min to Max (°F)	40 to 100
Power Adaptor Input (Voltage - Hertz - Current)	120V AC - 60 Hz - 0.35 A
Power Adaptor Output (Voltage - Current)	15V DC - 0.5 A
3 Volt Lithium Coin Cell Battery; Type	2032
Maximum Sulfate Concentration	265 mg/L



OUTLET SIDE VIEW



FRONT VIEW

GENERAL NOTES FOR ESTIMATING ONLY.
RESIN TANK DIAMETER AND HEIGHT FOR EACH MODEL ON NEXT PAGE.

SPECIFICATIONS

Model Number	BP150NJ
Service Flow (GPM)	6
Resin Tank Size; Diameter x Height (inches)	10 x 54
Resin (Cu. Ft.)	1.5
Brine Tank Size, Diameter x Height (inches)	16 x 33
Brine Tank Capacity (LBS NaCl)	258
Drain Line Flow Control (GPM)	1.7
Brine Line Re-Fill Flow Control (GPM)	0.5
Injector Color	1E - White

CYCLES* (Downflow Regenerant, Postfill)	
1st: Backwash (Minutes)	10
2nd: Regenerant Draw Down - Draw & Slow Rinse (Minutes)	50
3rd: Backwash (Minutes)	10
4th: Rinse - Fast Rinse (Minutes)	5
5th: Fill (LBS NaCl)	15
6th: End (Regeneration Complete)	-

Grains Capacity (see Note 1)	12,362
LBS NaCl Per Regeneration	15
(Re-) Fill (Minutes)	10
Gallons Water to Drain at 50 PSI Inlet Pressure (see Note 2)	66

* Factory Setting

1 - Grains Capacity is based on 8,241 grains per cubic foot as $CaCO_3$ operating capacity of Jacobi Resinex NR-1 for nitrate removal at 10 pounds of NaCl per cubic foot with an influent concentration of 600 mg/L TDS (sulfate concentration of 265 mg/L).

2 - Gallons to Drain = (Backwash Min. x DLFC GPM) + (Fast Rinse Min. x DLFC GPM) + (Regenerant Draw DN Min. x Slow Rinse GPM) + (LBS NaCl x 1 Gal./3 LBS NaCl)

Note: Slow Rinse GPM is obtained from injector charts for each color injector at various pressures.

PERFORMANCE DATA SHEET FOR MODEL BP150NJ

This system has been tested according to ASSE LEC 2008 for reduction of nitrate as detailed below. The concentration of nitrate entering the system was reduced to a concentration less than or equal to the EPA permissible MCL of 10 mg/L as specified in ASSE LEC 2008. **BP150NJ is certified by IAPMO R&T to NSF/ANSI 53, ASSE LEC 2008 and NSF/ANSI/CAN 372.**

Contaminant	Average Influent	Max Effluent	Avg % reduction	Average Effluent
Nitrate	30.17 mg/L	9.7 mg/L	79.96	6.0 mg/L

Nitrate reducing capacity between regenerations: 1766 gal
Rated Service Flow: 6 gpm
Rated Capacity: 1766 gallons
Working Pressure: 20 - 100 psi (1.4 - 7 kgf/cm²)
Working Temperature: 40 - 100 degF (4.4 - 37.7 degC)

While testing was performed under standard laboratory conditions, actual performance may vary.

LEAD FREE

UPC ®

PROTECT PLUS PRO, LLC
420 3RD AVENUE, NW
HICKORY, NC 28601
828.328.1142

Testing specifications
Flow Rate 6 gpm
Pressure 55 psi
Temperature 70.5 degF

Please refer to the Install Manual for general operation and maintenance requirements and the manufacturer's warranty.

EFFICIENCY OF NITRATE REMOVAL NORMALLY DEPENDS ON PRESENCE OF COMPETING ANIONS, ESPECIALLY SULFATES, WHICH CAN CAUSE THE DUMPING OF THE NITRATES. HOWEVER RESINEX ™ NR-1 ANION EXCHANGE RESIN USED IN THIS CONDITIONER HAS BEEN SPECIALLY DEVELOPED FOR SELECTIVE NITRATE REMOVAL FROM POTABLE WATER IN THE PRESENCE OF HIGH LEVELS OF SULFATE.

THUS THERE IS NO DANGER OF NITRATE DUMPING IN THIS NITRATE REMOVAL SYSTEM.

This system has been tested according to NSF/ANSI 53 for reduction of the substances listed below. The concentration of the indicated substances in water entering the system was reduced to a concentration less than or equal to the permissible limit for water leaving the system, as specified in NSF/ANSI 53.

Substance	Influent challenge concentration (mg/L)	Maximum permissible product water concentration (mg/L)
total PFAS[b]	0.00216 ± 20%	0.00002°

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NSF/ANSI Standard 53 PFAS Reduction PT 100%: Passed

Manufacturer's Name: Protect Plus Pro

Sample Type: R & D

Product: POE System

Flow Rate: 6 gpm (Control)

Filter Capacity: 1766 gallons

Cycle: 50/50

Conditioning Procedure: Follow Manufacturer's Instructions

Physical Description of Sample: POE System

Performance Indicator Device: N/A, test to 100% Capacity

Test Description: NSF/ANSI Std 53 PFAS Reduction Testing

Trade Designation/Model Number: BP150NJ

Unit Volume:

Performance Standard: NSF/ANSI Std 53 – 2022

Pass/Fail Criteria: PFOA+PFOS Combined Maximum Product Water Concentration 0.02 µg/L; PFNA 0.006 µg/L; PFHxS 0.02 µg/L; PFHpA 0.02 µg/L. Total PFAS maximum effluent concentration: 0.02 µg/L

Decision Rule: Simple Acceptance based on the NSF/ANSI Std 53.

Water Characteristics Run #1

Sample Point	pH (7.5±0.5)	Temperature (20±3°C)	Turbidity (<1 NTU)	TOC (1.0-2.0 mg/L)	TDS (200-500)	Flow Rate F1	Pressure F1
10 UV	7.46	20.1	0.5	1.4	249	6	60
20%	7.41	20.1	0.5	1.4	255	6	60
40%	7.40	20	0.4	1.4	254	6	60
60%	7.43	20.1	0.4	1.5	252	6	60
80%	7.41	20.2	0.5	1.5	255	6	60
100%	7.44	20.3	0.4	1.4	257	6	60
Average	**7.43**	**20.1**	**0.5**	**1.4**	**254**	**6**	**60**

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Water Characteristics Run #2

Sample Point	pH (7.5±0.5)	Temperature (20±3°C)	Turbidity (<1 NTU)	TOC (1.0-2.0 mg/L)	TDS (200-500)	Flow Rate F1	Pressure F1
10 UV	7.42	20.1	0.4	1.2	248	6	60
20%	7.46	20.3	0.4	1.2	249	6	60
40%	7.44	20.2	0.5	1.3	247	6	60
60%	7.45	20.1	0.4	1.2	246	6	60
80%	7.42	20.3	0.5	1.3	248	6	60
100%	7.41	20.1	0.5	1.2	249	6	60
Average	**7.43**	**20.1**	**0.5**	**1.4**	**254**	**6**	**60**

Water Characteristics Run #3

Sample Point	pH (7.5±0.5)	Temperature (20±3°C)	Turbidity (<1 NTU)	TOC (1.0-2.0 mg/L)	TDS (200-500)	Flow Rate F1	Pressure F1
10 UV	7.45	19.2	0.5	1.2	249	6	60
20%	7.42	19.4	0.5	1.1	252	6	60
40%	7.44	19.8	0.4	1.3	254	6	60
60%	7.41	20.2	0.5	1.2	248	6	60
80%	7.43	20.3	0.4	1.1	251	6	60
100%	7.41	20.1	0.5	1.2	253	6	60
Average	**7.43**	**20.1**	**0.5**	**1.4**	**254**	**6**	**60**

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PFOA Data Summary Table Run # 1

Sample Point	Accumulated 1 Volume Effluent	Influent 1 PFOA (µg/L)	Effluent PFOA Concentration (µg/L)
10 UV	10 UV	0.52	<0.01
20%	353.20 gallons	0.49	<0.01
40%	706.40 gallons	0.52	<0.01
60%	1059.60 gallons	0.50	<0.01
80%	1412.80 gallons	0.51	<0.01
100%	1766 gallons	0.52	<0.01

PFOA Data Summary Table Run # 2

Sample Point	Accumulated 2 Volume Effluent	Influent 2 PFOA (µg/L)	Effluent PFOA Concentration (µg/L)
10 UV	10 UV	0.49	0.01
20%	353.20 gallons	0.50	0.01
40%	706.40 gallons	0.48	0.01
60%	1059.60 gallons	0.49	0.01
80%	1412.80 gallons	0.51	0.01
100%	1766 gallons	0.50	0.01

PFOA Data Summary Table Run # 3

Sample Point	Accumulated 3 Volume Effluent	Influent 3 PFOA (µg/L)	Effluent PFOA Concentration (µg/L)
10 UV	10 UV	0.52	<0.01
20%	353.20 gallons	0.50	0.01
40%	706.40 gallons	0.51	0.01
60%	1059.60 gallons	0.51	0.01
80%	1412.80 gallons	0.50	0.01
100%	1766 gallons	0.50	0.01

PFOS Data Summary Table Run # 1

Sample Point	Accumulated 1 Volume Effluent	Influent 1 PFOS (µg/L)	Effluent PFOS Concentration (µg/L)
10 UV	10 UV	1.20	<0.01
20%	353.20 gallons	1.10	<0.01
40%	706.40 gallons	1.10	<0.01
60%	1059.60 gallons	1.10	0.01
80%	1412.80 gallons	1.20	0.01
100%	1766 gallons	1.00	0.01

PFOS Data Summary Table Run # 2

Sample Point	Accumulated 2 Volume Effluent	Influent 2 PFOS (µg/L)	Effluent PFOS Concentration (µg/L)
10 UV	10 UV	1.10	<0.01
20%	353.20 gallons	1.10	0.01
40%	706.40 gallons	1.10	<0.01
60%	1059.60 gallons	1.10	0.01
80%	1412.80 gallons	1.00	0.01
100%	1766 gallons	1.00	<0.01

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PFOS Data Summary Table Run # 3

Sample Point	Accumulated 3 Volume Effluent	Influent 3 PFOS (µg/L)	Effluent PFOS Concentration (µg/L)
10 UV	10 UV	1.10	0.01
20%	353.20 gallons	1.00	0.01
40%	706.40 gallons	1.10	<0.01
60%	1059.60 gallons	0.99	0.01
80%	1412.80 gallons	0.98	0.01
100%	1766 gallons	1.10	0.01

PFOA & PFOS Data Summary Run # 1

Sample Point	Accumulated Volume Effluent 1	Influent 1 Total PFOA + PFOS Concentration (µg/L)	Effluent 1 Total PFOA + PFOS Concentration (µg/L)
10 UV	10 UV	1.72	<0.01
20%	353.20 gallons	1.59	<0.01
40%	706.40 gallons	1.62	<0.01
60%	1059.60 gallons	1.60	0.01
80%	1412.80 gallons	1.71	0.01
100%	1766 gallons	1.52	0.01

PFOA & PFOS Data Summary Run # 2

Sample Point	Accumulated Volume Effluent 2	Influent 2 Total PFOA + PFOS Concentration (µg/L)	Effluent 2 Total PFOA + PFOS Concentration (µg/L)
10 UV	10 UV	1.59	0.01
20%	353.20 gallons	1.60	0.02
40%	706.40 gallons	1.58	0.01
60%	1059.60 gallons	1.59	0.02
80%	1412.80 gallons	1.51	0.02
100%	1766 gallons	1.50	0.01

PFOA & PFOS Data Summary Run # 3

Sample Point	Accumulated Volume Effluent 3	Influent 3 Total PFOA + PFOS Concentration (µg/L)	Effluent 3 Total PFOA + PFOS Concentration (µg/L)
10 UV	10 UV	1.62	0.01
20%	353.20 gallons	1.50	0.02
40%	706.40 gallons	1.61	0.01
60%	1059.60 gallons	1.50	0.02
80%	1412.80 gallons	1.48	0.02
100%	1766 gallons	1.60	0.02

PFNA Data Summary Table Run # 1

Sample Point	Accumulated Volume Effluent 1	Influent 1 PFNA (µg/L)	Effluent 1 PFNA Concentration (µg/L)
10 UV	10 UV	0.05	<0.01
20%	353.20 gallons	0.05	<0.01
40%	706.40 gallons	0.05	<0.01
60%	1059.60 gallons	0.05	<0.01
80%	1412.80 gallons	0.05	<0.01
100%	1766 gallons	0.05	<0.01

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PFNA Data Summary Table Run # 2

Sample Point	Accumulated Volume Effluent 2	Influent 2 PFNA (µg/L)	Effluent 2 PFNA Concentration (µg/L)
10 UV	10 UV	0.05	<0.01
20%	353.20 gallons	0.05	<0.01
40%	706.40 gallons	0.05	<0.01
60%	1059.60 gallons	0.05	<0.01
80%	1412.80 gallons	0.05	<0.01
100%	1766 gallons	0.05	<0.01

PFNA Data Summary Table Run # 3

Sample Point	Accumulated Volume Effluent 3	Influent 3 PFNA (µg/L)	Effluent 3 PFNA Concentration (µg/L)
10 UV	10 UV	0.05	<0.01
20%	353.20 gallons	0.05	<0.01
40%	706.40 gallons	0.05	<0.01
60%	1059.60 gallons	0.05	<0.01
80%	1412.80 gallons	0.05	<0.01
100%	1766 gallons	0.05	<0.01

PFHxS Data Summary Table Run # 1

Sample Point	Accumulated Volume Effluent 1	Influent 1 PFHxS (µg/L)	Effluent 1 PFHxS Concentration (µg/L)
10 UV	10 UV	0.32	<0.01
20%	353.20 gallons	0.32	<0.01
40%	706.40 gallons	0.32	<0.01
60%	1059.60 gallons	0.30	<0.01
80%	1412.80 gallons	0.32	<0.01
100%	1766 gallons	0.32	<0.01

PFHxS Data Summary Table Run # 2

Sample Point	Accumulated Volume Effluent 2	Influent 2 PFHxS (µg/L)	Effluent 2 PFHxS Concentration (µg/L)
10 UV	10 UV	0.31	<0.01
20%	353.20 gallons	0.31	<0.01
40%	706.40 gallons	0.30	<0.01
60%	1059.60 gallons	0.31	<0.01
80%	1412.80 gallons	0.31	<0.01
100%	1766 gallons	0.31	<0.01

PFHxS Data Summary Table Run # 3

Sample Point	Accumulated Volume Effluent 3	Influent 3 PFHxS (µg/L)	Effluent 3 PFHxS Concentration (µg/L)
10 UV	10 UV	0.31	<0.01
20%	353.20 gallons	0.30	<0.01
40%	706.40 gallons	0.32	<0.01
60%	1059.60 gallons	0.30	<0.01
80%	1412.80 gallons	0.28	<0.01
100%	1766 gallons	0.29	<0.01

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PFHpA Data Summary Table Run # 1

Sample Point	Accumulated Volume Effluent 1	Influent 1 PFHpA (µg/L)	Effluent 1 PFHpA Concentration (µg/L)
10 UV	10 UV	0.04	<0.01
20%	353.20 gallons	0.04	0.01
40%	706.40 gallons	0.04	0.01
60%	1059.60 gallons	0.04	<0.01
80%	1412.80 gallons	0.04	0.01
100%	1766 gallons	0.04	0.01

PFHpA Data Summary Table Run # 2

Sample Point	Accumulated Volume Effluent 2	Influent 2 PFHpA (µg/L)	Effluent 2 PFHpA Concentration (µg/L)
10 UV	10 UV	0.04	<0.01
20%	353.20 gallons	0.04	<0.01
40%	706.40 gallons	0.04	<0.01
60%	1059.60 gallons	0.04	<0.01
80%	1412.80 gallons	0.04	<0.01
100%	1766 gallons	0.04	<0.01

PFHpA Data Summary Table Run # 3

Sample Point	Accumulated Volume Effluent 3	Influent 3 PFHpA (µg/L)	Effluent 3 PFHpA Concentration (µg/L)
10 UV	10 UV	0.04	<0.01
20%	353.20 gallons	0.04	<0.01
40%	706.40 gallons	0.04	<0.01
60%	1059.60 gallons	0.04	<0.01
80%	1412.80 gallons	0.04	<0.01
100%	1766 gallons	0.04	<0.01

PFBS Data Summary Table Run # 1

Sample Point	Accumulated Volume Effluent 1	Influent 1 PFBS (µg/L)
10 UV	10 UV	0.26
20%	353.20 gallons	0.25
40%	706.40 gallons	0.26
60%	1059.60 gallons	0.25
80%	1412.80 gallons	0.25
100%	1766 gallons	0.27

PFBS Data Summary Table Run # 2

Sample Point	Accumulated Volume Effluent 2	Influent 2 PFBS (µg/L)
10 UV	10 UV	0.25
20%	353.20 gallons	0.27
40%	706.40 gallons	0.26
60%	1059.60 gallons	0.27
80%	1412.80 gallons	0.26
100%	1766 gallons	0.25

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PFBS Data Summary Table Run # 3

Sample Point	Accumulated Volume Effluent 3	Influent 3 PFBS (µg/L)
10 UV	10 UV	0.27
20%	353.20 gallons	0.27
40%	706.40 gallons	0.27
60%	1059.60 gallons	0.26
80%	1412.80 gallons	0.27
100%	1766 gallons	0.26

PFDA Data Summary Table Run # 1

Sample Point	Accumulated Volume Effluent 1	Influent 1 PFDA (µg/L)
10 UV	10 UV	0.01
20%	353.20 gallons	0.01
40%	706.40 gallons	0.01
60%	1059.60 gallons	0.01
80%	1412.80 gallons	0.01
100%	1766 gallons	0.01

PFDA Data Summary Table Run # 2

Sample Point	Accumulated Volume Effluent 2	Influent 2 PFDA (µg/L)
10 UV	10 UV	0.01
20%	353.20 gallons	0.01
40%	706.40 gallons	0.01
60%	1059.60 gallons	0.01
80%	1412.80 gallons	0.01
100%	1766 gallons	0.01

PFDA Data Summary Table Run # 3

Sample Point	Accumulated Volume Effluent 3	Influent 3 PFDA (µg/L)
10 UV	10 UV	0.01
20%	353.20 gallons	0.01
40%	706.40 gallons	0.01
60%	1059.60 gallons	0.01
80%	1412.80 gallons	0.01
100%	1766 gallons	0.01

Total PFAS Data Summary Table Run # 1

Sample Point	Accumulated Volume Effluent 1	Influent 1 PFAS (µg/L)	Effluent 1 PFAS Concentration (µg/L)	Passing Criteria
10 UV	10 UV	2.40	<0.01	Passed
20%	353.20 gallons	2.26	0.01	Passed
40%	706.40 gallons	2.30	0.01	Passed
60%	1059.60 gallons	2.25	0.01	Passed
80%	1412.80 gallons	2.38	0.02	Passed
100%	1766 gallons	2.21	0.02	Passed

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Total PFAS Data Summary Table Run # 2

Sample Point	Accumulated Volume Effluent 1	Influent 2 PFAS (µg/L)	Effluent 2 PFAS Concentration (µg/L)	Passing Criteria
10 UV	10 UV	2.25	0.01	Passed
20%	353.20 gallons	2.28	0.02	Passed
40%	706.40 gallons	2.24	0.01	Passed
60%	1059.60 gallons	2.27	0.02	Passed
80%	1412.80 gallons	2.18	0.02	Passed
100%	1766 gallons	2.16	0.01	Passed

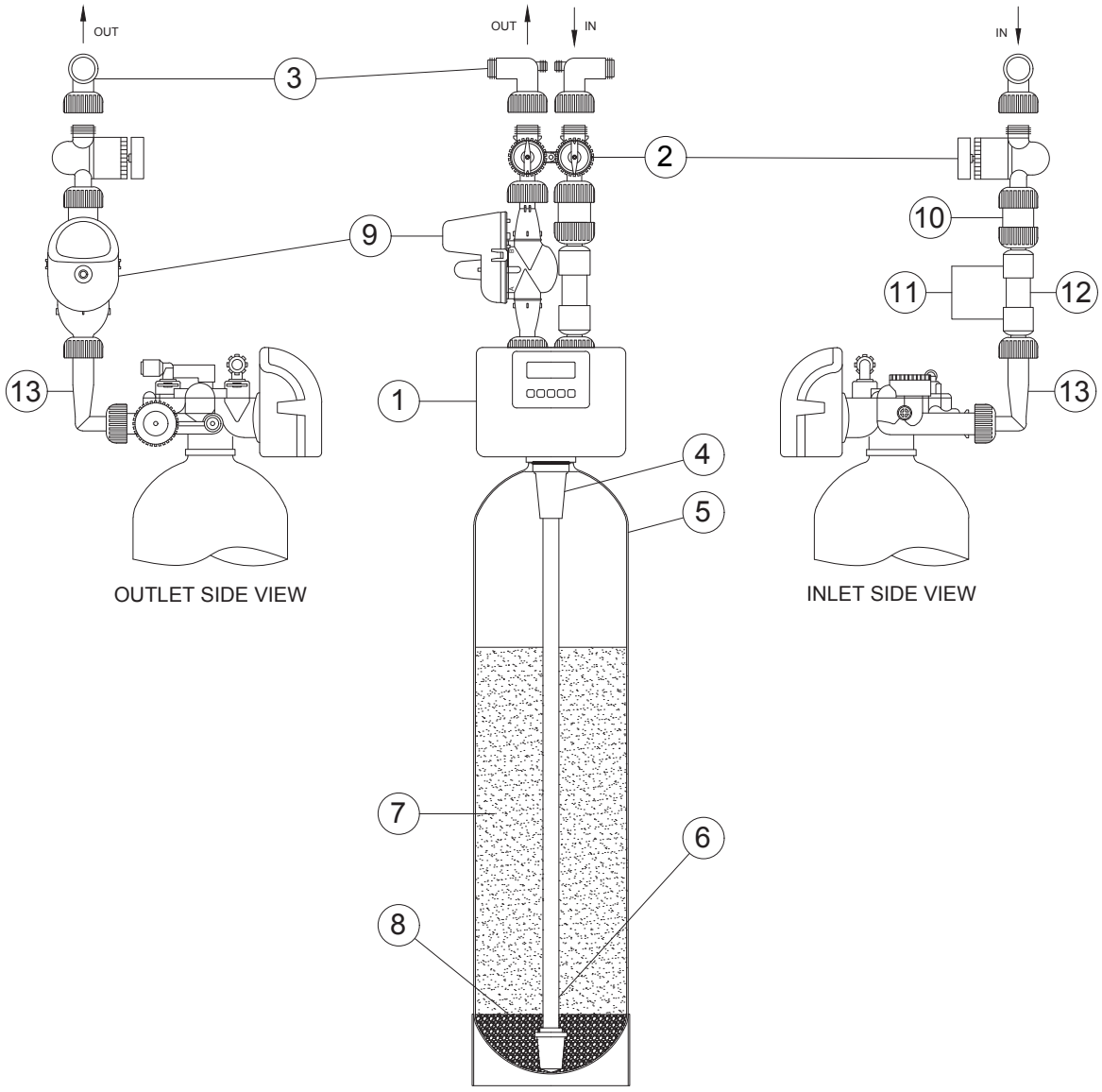
Total PFAS Data Summary Table Run # 3

Sample Point	Accumulated Volume Effluent 3	Influent 3 PFAS (µg/L)	Effluent 3 PFAS Concentration (µg/L)	Passing Criteria
10 UV	10 UV	2.30	0.01	Passed
20%	353.20 gallons	2.17	0.02	Passed
40%	706.40 gallons	2.30	0.01	Passed
60%	1059.60 gallons	2.16	0.02	Passed
80%	1412.80 gallons	2.13	0.02	Passed
100%	1766 gallons	2.25	0.02	Passed

UNIT ASSEMBLY

BRITA PRO NITRATE REMOVAL SYSTEM
MODEL BP150NJ

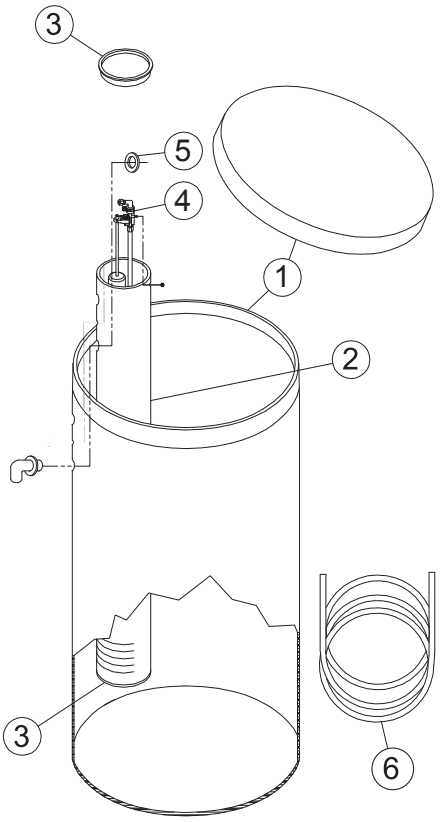
NOTE: J = JACKET (blow molded tank jacket); jacket over mineral tank not shown on this diagram



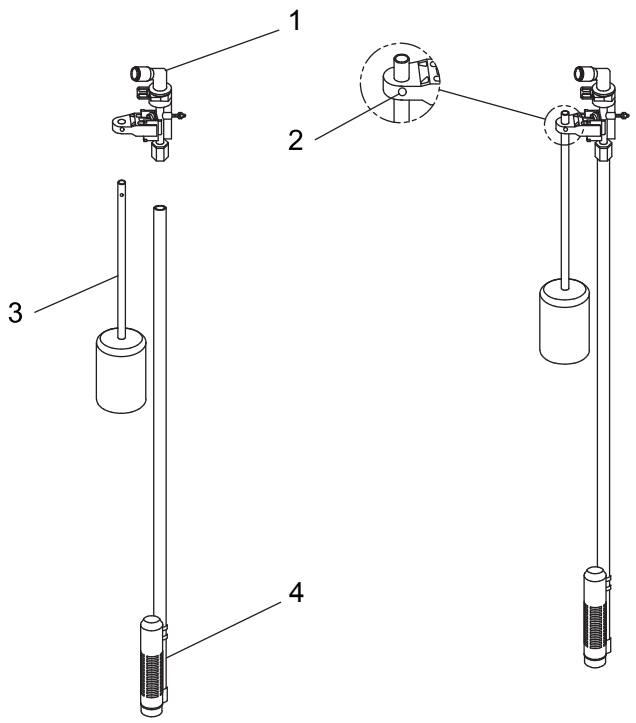
Dwg. No.	Order No.	Description	Qty
1	BPWS1LR & BPVLVCOVER (detailed components shown in this manual)	Control Valve & Face Plate	1
2	BP2000	Bypass Valve	1
3	BPV3007	Fitting 1” PVC Male NPT Elbow Assembly (2 per bag)	2
4	BPD1203	Top Distributor	1
5	BPFG1054J	Tank	1
6	BP912-55 (cut for 54 in.)	Basket DIstributor Assembly	1
7	BPA4103	Nitrate Removal Resin (66 lbs. 1.5 cu.ft.)	1
8	BPA8072-15	Support Bedding (15 lbs.)	1
9	BPV3070FM	No Nitrate Water Bypass 1/125 F-M	1
10	BPV3467	Bypass In-line Adapter/Spacer	1
11	BPV300707	Fitting 1-1/4 Plastic Male Solvent Assembly (2 per bag)	2
12	BPSCH80	1-1/4” x 4-1/8” PVC SCH 80 Pipe	1
13	BPV319101	Fitting Vertical Adapter Assembly (2 per bag)	2

BRINE TANK ASSEMBLY

Dwg. No.	Order No.	Description	Qty
1	BPJ7057	16x33 Assembly	1
2	BPH1031	4x30 Slotted Brine Well	1
3	BPH7016	4 inch Brine Well Cap	2
4	BP4740	3/8” Brine Valve Assembly (using 1/2”x25” long air check assy.)	1
5	BPH1018	Two Piece Overflow Set	1
6	BPH1023	3/8”O.D.x6 Ft. Poly Tubing	1



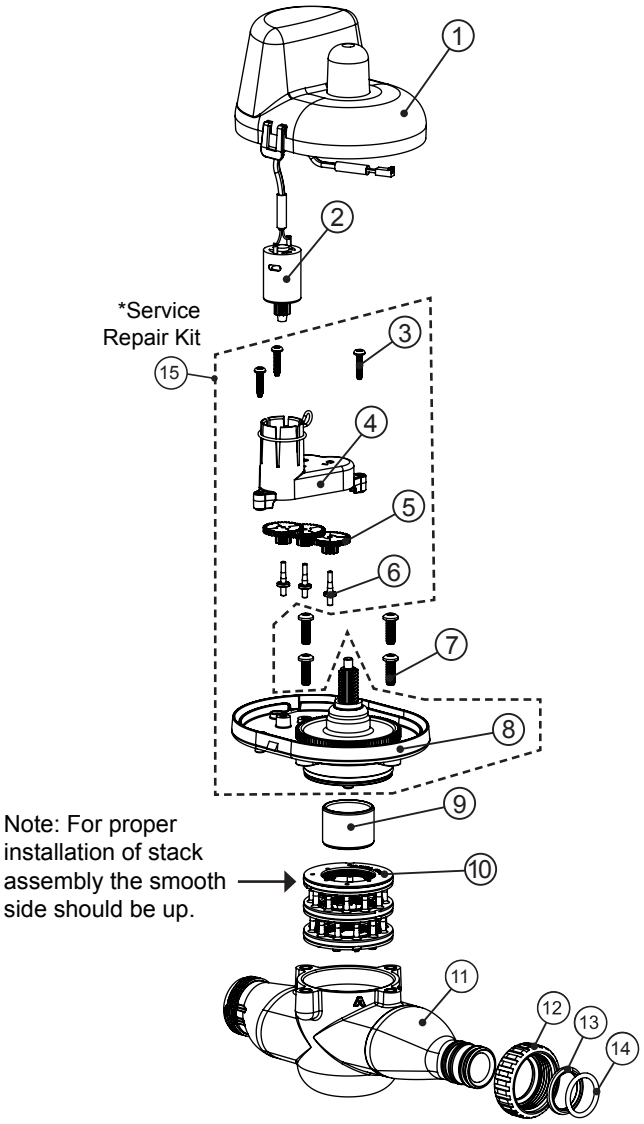
BRINE VALVE ASSEMBLY



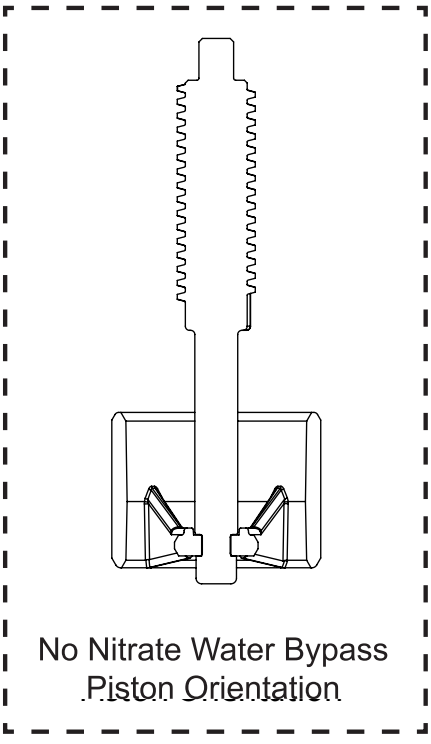
Dwg No.	Order No.	Description	Qty
1	BPH4600	3/8” Safety Brine Valve	1
2	BP10151	Pin	1
3	BPH46409.5	Float Assembly	1
4	BPH450048	Air Check Assembly (uncut length) see brine tank assy for cut length	1

NO HARDWATER BYPASS
(Order No. BPV3070FM)

Dwg. No.	Order No.	Description	Qty
1	BPV3073	MAV/NOHWBY COVER ASY	1
2	BPV3476	MOTOR ASY 8 FT	1
3	BPV3592	SCREW #8-3/4 PHPN T-25 SS	3
4	BPV326201	ALT/2BY REDUCING GEAR CVR ASSEMBLY	1
5	BPV311001	DRIVE REDUCING GEAR PLAIN	3
6	BPV3264	BYPASS REDUCTION GEAR AXLE	3
7	BPV3527	SCREW 1/4-20 X 3/4 BHSCS SS	4
8	BPV3072	MAV/NHWB 1/125/15 DRIVE ASY	1
9	BPV350601	MAV/NHWB 1/125/15 PISTON	1
10	BPV3074	MAV 1/125/15 STACK ASY	1
11	BPV3521FM	NO NITRATE WATER BYPASS BODY ASSEMBLY F-M	1
12	BPV3151	NUT 1 QC	1
13	BPV3150	SPLIT RING	1
14	BPV3105	O-RING 215	1
15*	V3042	MAV & NHWB SERVICE PACK	



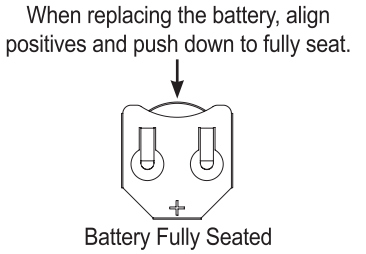
•Operating Pressures:
20 PSI Minimum / 125 PSI Maximum
•Operating Temperatures:
40°F Minimum / 110°F Maximum

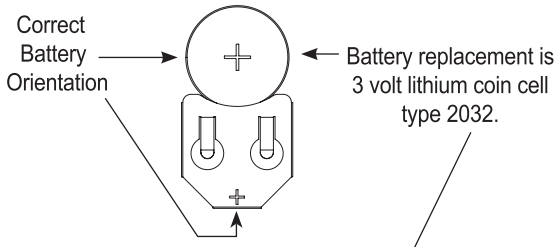


No Nitrate Water Bypass
Piston Orientation

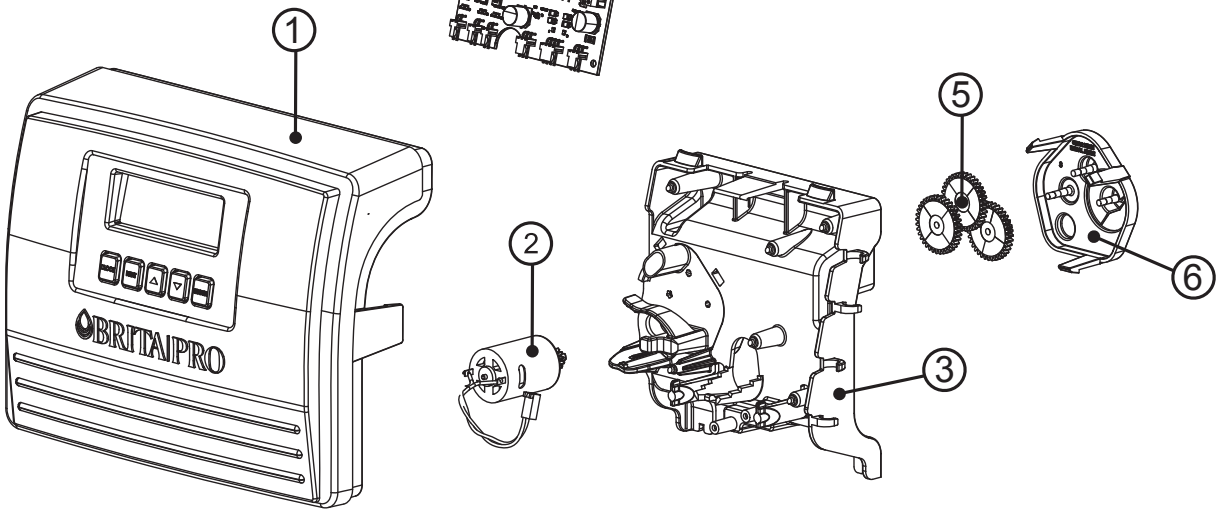
FRONT COVER AND DRIVE ASSEMBLY

Drawing No.	Order No.	Description	Quantity
1	BPVLVCOVER	COVER ASSEMBLY FACE PLATE	
2	BPV310701	MOTOR	1
3	BPV3002A	DRIVE BRACKET & SPRING CLIP ASSEMBLY	1
4	BPV4478LR	PC BOARD	1
5	BPV3110	DRIVE GEAR 12X36	3
6	BPV3109	DRIVE GEAR COVER	1





AC Adapter Order No.	(Not Shown) V3186-06
Supply Voltage	100-120 VAC
Supply Frequency	50/60 Hz
Output Voltage	15 VDC
Output Current	500 mA

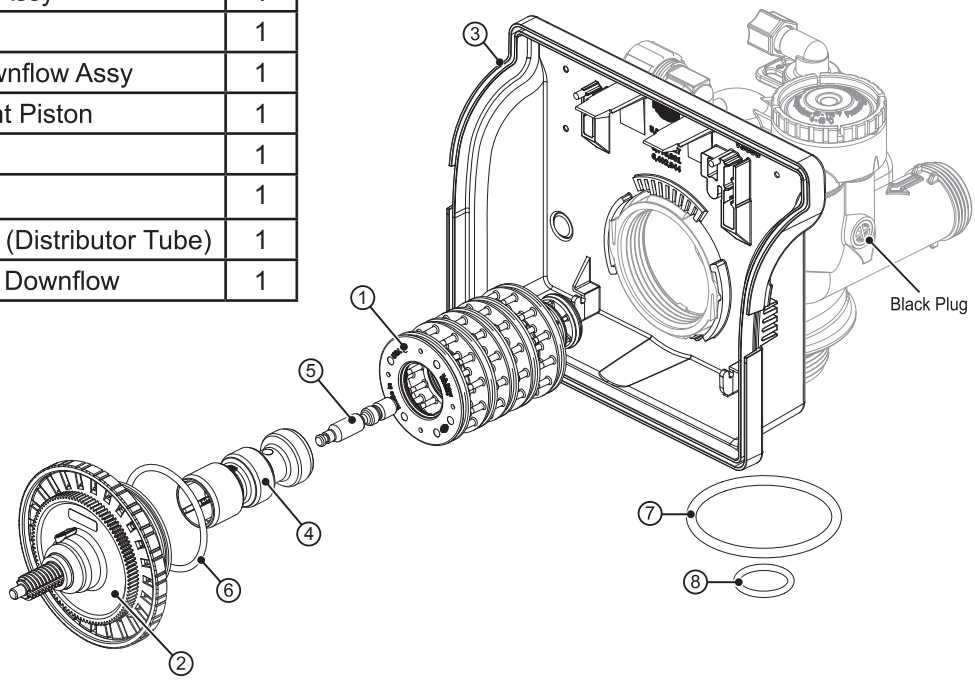


After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons, unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds. This resets the electronics and establishes the service piston position. The display should flash the software version and then reset the valve to the service position.

DRIVE CAP ASSEMBLY, DOWNFLOW PISTON, REGENERANT PISTON AND SPACER STACK ASSEMBLY

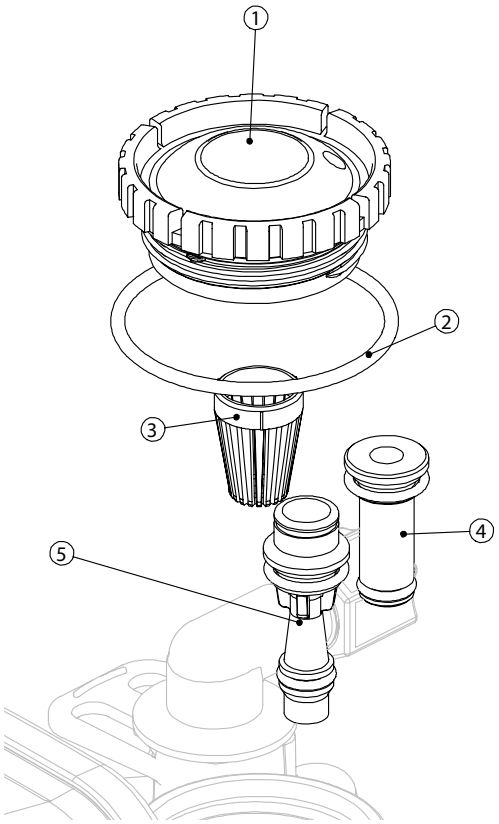
Drawing No.	Order No.	Description	Qty
1	BPV3005	Spacer Stack Assembly	1
2	BPV3004	Drive Cap Assy	1
3	BPV3178LP	Back Plate	1
4	BPV3011	Piston Downflow Assy	1
5	BPV3174	Regenerant Piston	1
6	BPV3135	O-ring 228	1
7	BPV3180	O-ring 337	1
8	BPV3105	O-ring 215 (Distributor Tube)	1
Not Shown	BPV3001	Body Assy Downflow	1

Do not use Vaseline, oils, other hydrocarbon lubricants or spray silicone anywhere. A silicone lubricant may be used on black o-rings. **Avoid any type of lubricants, including silicone, on the clear lip seals.**



After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons, unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds. This resets the electronics and establishes the service piston position. The display should flash the software version and then reset the valve to the service position.

INJECTOR CAP, INJECTOR SCREEN, INJECTOR, PLUG AND O-RING



Drawing No.	Order No.	Description	Qty
1	BPV3176	INJECTOR CAP	1
2	BPV3152	O-RING 135	1
3	BPV317701	INJECTOR SCREEN CAGE	1
4	BPV30101Z	INJECTOR ASSY Z PLUG	1
5	BPV30101A	INJECTOR ASSY A BLACK	1
	BPV30101B	INJECTOR ASSY B BROWN	
	BPV30101C	INJECTOR ASSY C VIOLET	
	BPV30101D	INJECTOR ASSY D RED	
	BPV30101E	INJECTOR ASSY E WHITE	
	BPV30101F	INJECTOR ASSY F BLUE	
	BPV30101G	INJECTOR ASSY G YELLOW	
	BPV30101H	INJECTOR ASSY H GREEN	
	BPV30101I	INJECTOR ASSY I ORANGE	
	BPV30101J	INJECTOR ASSY J LIGHT BLUE	
	BPV30101K	INJECTOR ASSY K LIGHT GREEN	
Not Shown	BPV3170	O-RING 011	*
Not Shown	BPV3171	O-RING 013	*

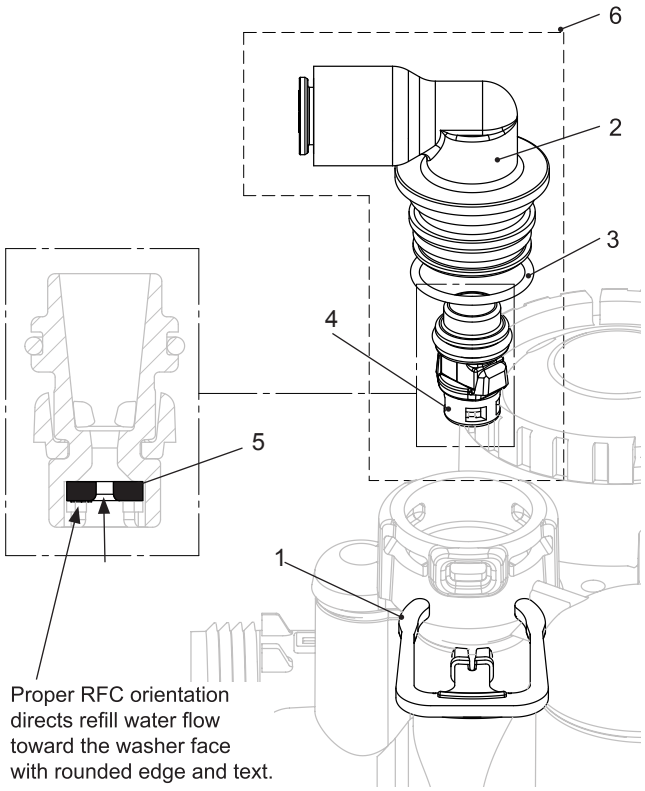
* The injector plug and the injector each contain one 011 (lower) and 013 (upper) o-ring.

The nut and caps are designed to be unscrewed or tightened by hand or with the service spanner wrench (see page 26). If necessary a pliers can be used to unscrew the nut or cap. Do not use a pipe wrench to tighten or loosen nuts or caps. Do not place a screwdriver in slots on caps and/or tap with a hammer.

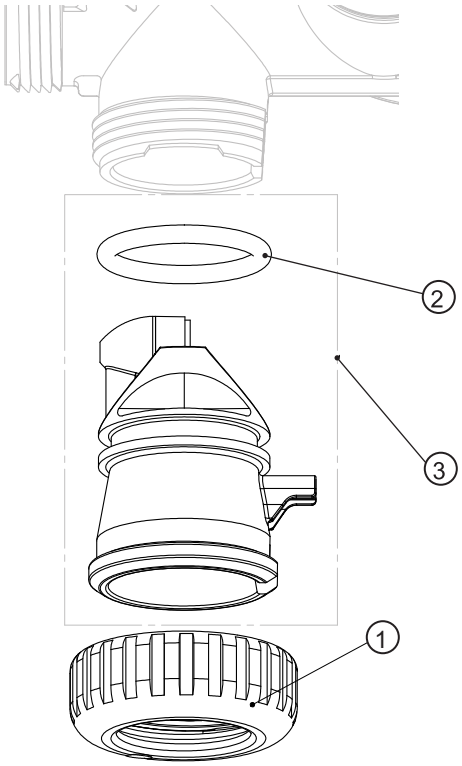
REFILL FLOW CONTROL ASSEMBLY

Drawing No.	Order No.	Description	Qty
1	BPH4615	Elbow Locking Clip	1
2	BPH4628	Elbow 3/8” Liquifit	1
3	BPV3163	O-ring 019	1
4	BPV316501*	RFC Retainer Assy (0.5 gpm)	1
5	BPV3182	RFC	1
6	BPV414401	Elbow 3/8 Liquifit Assy w/RFC	1

*Assembly includes BPV3182 RFC.



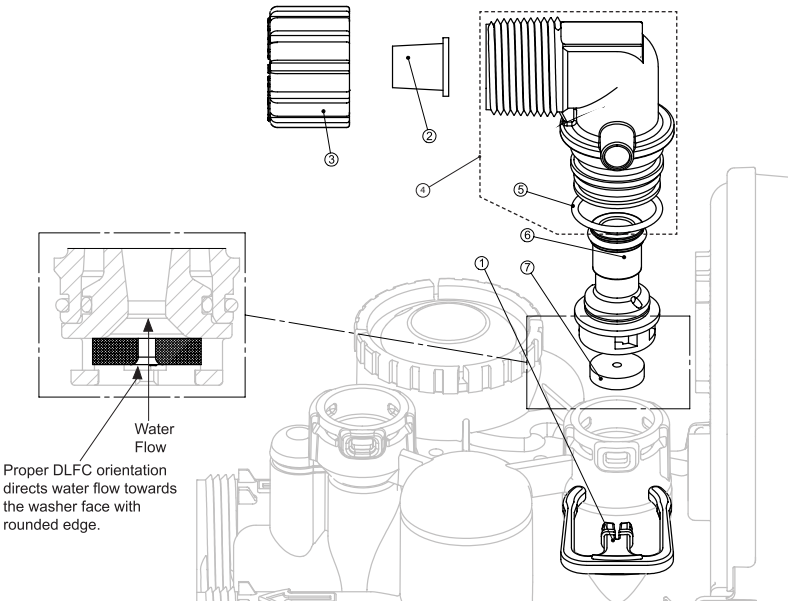
METER PLUG



Drawing No.	Order No.	Description	Qty.
1	BPV3151	Nut 1” QC	1
2	BPV3105	O-Ring 215	1
3	BPV3003-01	Meter Plug Assembly	1

DRAIN LINE - 3/4”

Dwg No.	Order No.	Description	Qty
1	BPH4615	Elbow Locking Clip	1
2	BPV3194	Polytube insert 5/8	Option
3	BPV3192	Nut ¾ Drain Elbow	Option
4*	BPV3158	Drain Elbow ¾ Male	1
5	BPV3163	O-ring 019	1
6*	BPV3159	DLFC Retainer Assy	1
7	BPV3162007	DLFC 0.7 gpm for ¾	One DLFC must be used if ¾ fitting is used
	BPV3162010	DLFC 1.0 gpm for ¾	
	BPV3162013	DLFC 1.3 gpm for ¾	
	BPV3162017	DLFC 1.7 gpm for ¾	
	BPV3162022	DLFC 2.2 gpm for ¾	
	BPV3162027	DLFC 2.7 gpm for ¾	
	BPV3162032	DLFC 3.2 gpm for ¾	
	BPV3162042	DLFC 4.2 gpm for ¾	
	BPV3162053	DLFC 5.3 gpm for ¾	
	BPV3162065	DLFC 6.5 gpm for ¾	
	BPV3162075	DLFC 7.5 gpm for ¾	
	BPV3162090	DLFC 9.0 gpm for ¾	
	BPV3162100	DLFC 10.0 gpm for ¾	



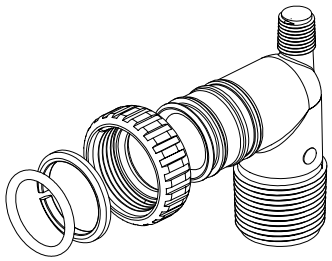
*4 and 6 can be ordered as a complete assembly - BPV3331 Drain Elbow and Retainer Assy

Do not use pipe dope or other sealants on threads. Teflon tape must be used on threads of 3/4” NPT connection (unless using 5/8” polytubing).

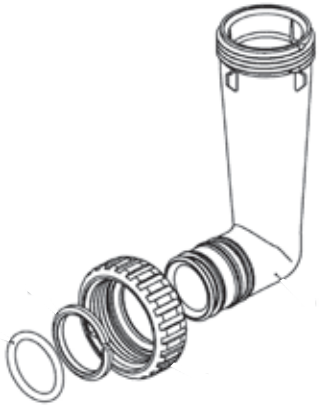
Do not use Vaseline, oils, other hydrocarbon lubricants or spray silicone anywhere. A silicone lubricant may be used on black o-rings.

INSTALLATION FITTING ASSEMBLIES

Order No: **BPV3007** (Standard)
Description: **Fitting 1” PVC Male NPT Elbow Assembly**



Order No. **BPV319101** (Standard)
Description: **Fitting Vertical Adapter Assembly**



Do not use pipe dope or other sealants on threads. Teflon tape must be used on threads of 1” NPT connection. Teflon tape is not necessary on the nut connection nor caps because of o-ring seals.

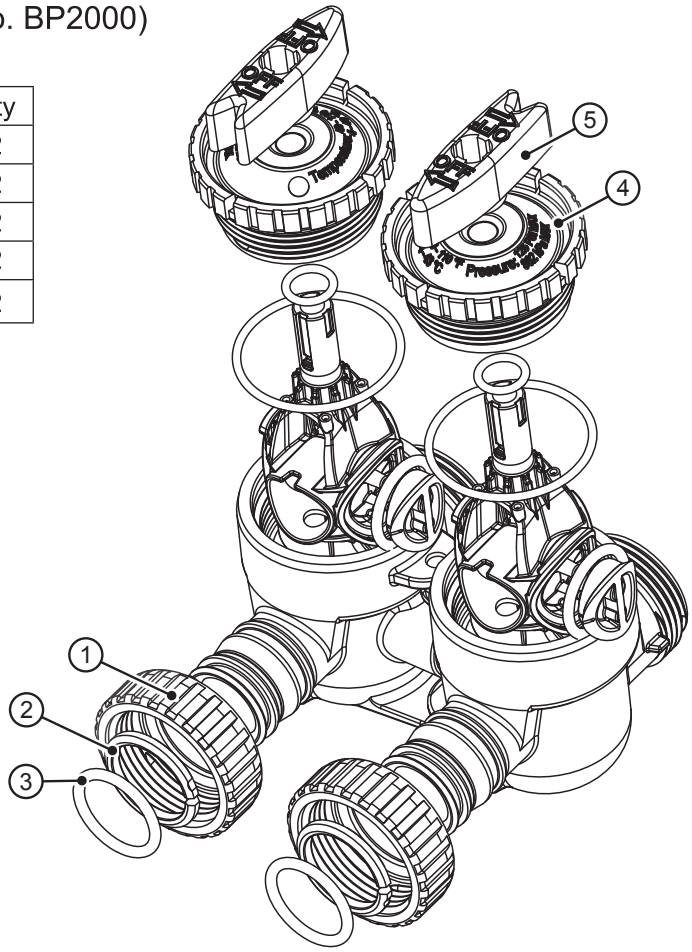
The nut and caps are designed to be unscrewed or tightened by hand or with the service spanner wrench. If necessary a pliers can be used to unscrew the nut or cap. Do not use a pipe wrench to tighten or loosen nuts or caps. Do not place a screwdriver in slots on caps and/or tap with a hammer.

Do not use Vaseline, oils, other hydrocarbon lubricants or spray silicone anywhere. A silicone lubricant may be used on black o-rings but is not necessary.

BYPASS VALVE

(Order No. BP2000)

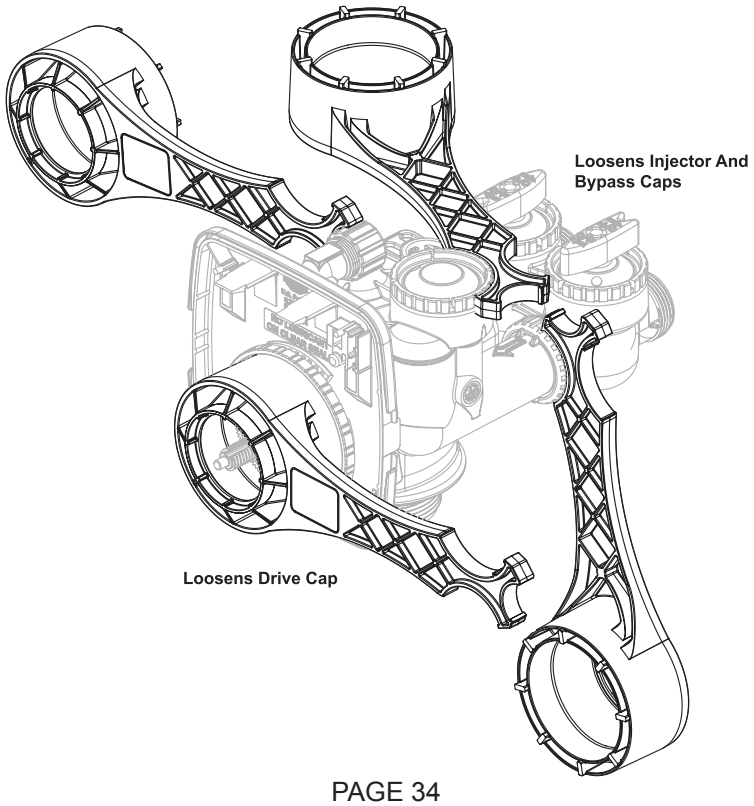
Drawing No.	Order No.	Description	Qty
1	BPV3151	Nut 1” Quick Connect	2
2	BPV3150	Split Ring	2
3	BPV3105	O-Ring 215	2
4	BPV3146	Bypass Cap	2
5	BPV3147	Bypass Handle	2



SERVICE SPANNER WRENCH

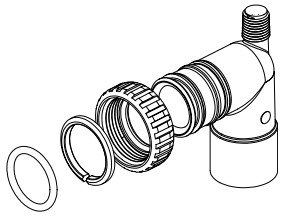
(Order No. BPV3193)

Although no tools are necessary to assemble or disassemble the valve, the wrench (shown in various positions on the valve) may be purchased to aid in assembly or disassembly.

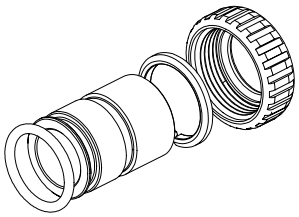


OPTIONAL INSTALLATION FITTING ASSEMBLIES

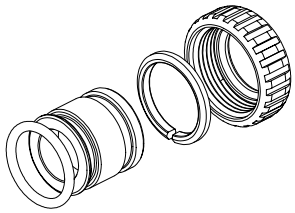
Order No: **BPV300701**
Description: **Fitting ¾" & 1" PVC Solvent 90° Assembly**



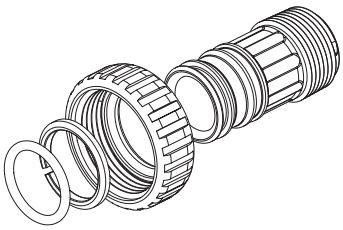
Order No: **BPV300702**
Description: **Fitting 1" Brass Sweat Assembly**
Do not install in California.



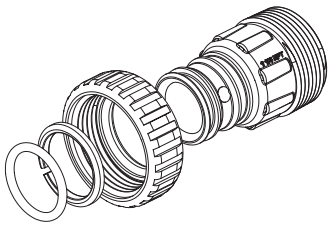
Order No: **BPV300703**
Description: **Fitting ¾" Brass Sweat Assembly**
Do not install in California.



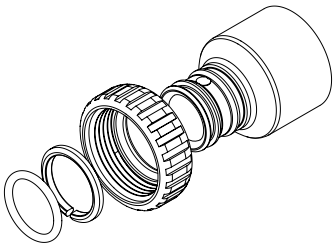
Order No: **BPV300704**
Description: **Fitting 1" Plastic Male NPT Assembly**



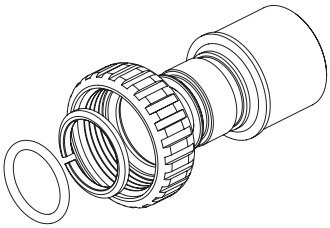
Order No: **BPV300705**
Description: **Fitting 1-¼" Plastic Male NPT Assembly**



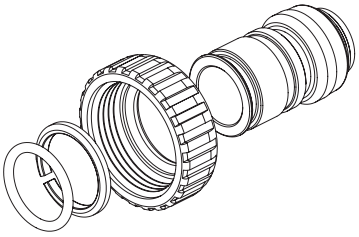
Order No. **BPV300707**
Description: **Fitting 1¼" & 1½" PVC Solvent Assembly**



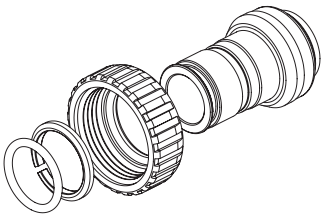
Order No: **BPV300709**
Description: **Fitting 1¼" & 1½" Brass Sweat Assembly**



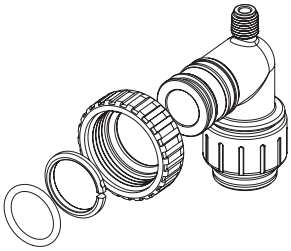
Order No. **BPV300712**
Description: **Fitting 3/4" Brass SharkBite Assembly**



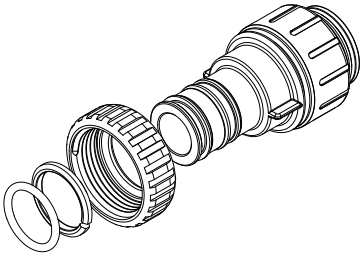
Order No. **BPV300713**
Description: **Fitting 1» Brass SharkBite Assembly**



Order No: **BPV300715**
Description: **WS1 FTG 3/4 JG QC 90 Assembly**



Order No: **BPV300717**
Description: **WS1 FTG 1" JG QC Assembly**



SERVICE INSTRUCTIONS



When servicing the valve, water may leak from the valve. Water from the valve may create a slip hazard. Clean up water spills.



Disconnect from electrical power prior to servicing the valve.

DRIVE ASSEMBLY:

Remove the valve cover to access the drive assembly.

Disconnect the power source plug (black wire) from the PC board prior to disconnecting the motor or water meter plugs from the PC board. The power source plug connects to the four-pin jack. The motor plug connects to the two-pin jack on the left-hand side of the PC board. The water meter plug (gray wire) connects to the three-pin jack on the far right-hand side of the PC board.

The PC board can be removed separately from the drive bracket but it is not recommended. Do not attempt to remove the display panel from the PC board. Handle the board by the edges. To remove the PC board from the drive bracket, unplug the power, water meter and motor plugs from the PC board. Lift the middle latch along the top of the drive bracket while pulling outward on the top of the PC board. The drive bracket has two plastic pins that fit into the holes on the lower edge of the PC board. Once the PC board is tilted about 45° from the drive bracket it can be lifted off of these pins. To reinstall the PC board, position the lower edge of the PC board so that the holes in the PC board line up with the plastic pins. Push the top of the PC board towards the valve until it snaps under the middle latch, weave the power and water meter wires into the holders and reconnect the motor, water meter and power plugs.

The drive bracket must be removed to access the drive cap assembly and pistons or the drive gear cover. It is not necessary to remove the PC board from the drive bracket to remove the drive bracket. To remove the drive bracket start by removing the plugs for the power source and the water meter. Unweave the wires from the side holders. Two tabs on the top of the drive back plate hold the drive bracket in place. Simultaneously lift the two tabs and gently ease the top of the drive bracket forward. The lower edge of the drive bracket has two notches that rest on the drive back plate. Lift up and outward on the drive bracket to disengage the notches.

To reassemble, seat the bottom of the drive bracket so the notches are engaged at the bottom of the drive back plate. Push the top of the drive bracket toward the two latches. The drive bracket may have to be lifted slightly to let the threaded piston rod pass through the hole in the drive bracket. Maintain a slight engaging force on top of the drive bracket while deflecting the bracket slightly to the left by pressing on the side of the upper right corner. This helps the drive gears mesh with the drive cap assembly. The drive bracket is properly seated when it snaps under the latches on the drive back plate. If resistance is felt before latching, then notches are not fully engaged, the piston rod is not in hole, the wires are jammed between the drive bracket and drive back plate, or the gear is not engaging the drive cap assembly.

To inspect the drive gears, the drive gear cover needs to be removed. Before trying to remove the gear cover, the drive bracket must be removed from the drive back plate. (Refer to the instructions above regarding removing the drive bracket from the drive back plate. The drive gear cover can be removed from the drive bracket without removing the motor or the PC board.) The drive gear cover is held in place on the drive bracket by three clips. The largest of the three clips is always orientated to the bottom of the drive bracket. With the PC board facing up, push in and down on the large clip on the drive gear cover. Handle the cover and the gears carefully so that the gears do not fall off the pegs in the cover.

Replace broken or damaged drive gears. Do not lubricate any of the gears. Avoid getting any foreign matter on the reflective coating because dirt or oils may interfere with pulse counting.

The drive gear cover only fits on one way, with the large clip orientated towards the bottom. If all three clips are outside of the gear shroud on the drive bracket the drive gear cover slips easily into place.

The drive bracket does not need to be removed from the drive plate if the motor needs to be removed. To remove the motor, disconnect the power and motor plugs from the jacks on the PC board. Move the spring clip loop to the right and hold. Rotate the motor at least a ¼ turn in either direction so the wires are vertical (up & down) before gently pulling on the wire connectors to remove the motor. Pulling directly on the wires without rotating the motor may break the wires off the motor.

Replace the motor if necessary. Do not lubricate the motor or the gears. To reinstall the motor, move the spring clip loop to the right and hold. Gently turn the motor while inserting so that the gear on the motor meshes with the gears under the drive gear cover. Release the spring clip loop and continue to rotate the motor until the wires are horizontal and the motor housing engages the small plastic bulge inside the drive bracket motor retainer. Reconnect the motor plug to the two-pronged jack on the lower left side of the PC board. If the motor will not easily engage with the drive gears when reinstalling, lift and slightly rotate the motor before reinserting. Reconnect the power plug.

Replace the valve cover. After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds.

This resets the electronics and establishes the service piston position. The display should flash all wording, then flash the software version and then reset the valve to the service position.

DRIVE CAP ASSEMBLY, MAIN PISTON AND REGENERANT PISTON:

The drive assembly must be removed to access the drive cap assembly. The drive cap assembly must be removed to access the piston(s). The drive cap assembly is threaded into the control valve body and seals with an o-ring. To remove the drive cap assembly use the clack service wrench.

The drive cap assembly contains the drive cap, the main drive gear, drive cap spline, piston rod and various other parts that should not be dissembled in the field. The only replaceable part on the drive cap assembly is the o-ring. Attached to the drive cap assembly is the main piston and if a regenerant is used, a regenerant piston.

The regenerant piston (the small diameter one behind the main piston) is removed from the main piston by pressing sideways and unsnapping it from its latch. Chemically clean in dilute sodium bisulfite or vinegar, or replace the regenerant piston if needed. To remove the main piston fully extend the piston rod and then unsnap the main piston from its latch by pressing on the side with the number. Chemically clean in dilute sodium bisulfite or vinegar, or replace the main piston.

Reattach the main piston to the drive cap assembly. Reattach the regenerant piston (if needed) to the main piston. Reinsert the drive cap assembly and piston into the spacer stack assembly and hand tighten the drive cap assembly. Continue to tighten the drive cap assembly using a screwdriver as a ratchet until the black o-ring on the spacer stack assembly is no longer visible through the drain port. Excessive force can break the notches molded into the drive back plate. Make certain that the main drive gear still turns freely.

The exact position of the piston is not important as long as the main drive gear turns freely. Reattach the drive assembly to the control valve and connect all plugs. After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds.
This resets the electronics and establishes the service piston position. The display should flash all wording, then flash the software version and then reset the valve to the service position.

REFILL FLOW CONTROL ASSEMBLY:

To clean or replace the refill flow control, pull out the elbow-locking clip and then pull straight up on the elbow. Replace the elbow locking clip in the slot so that it is not misplaced. Twist to remove the white flow control retainer. The flow control can be removed by prying upward through the side slots of the retainer with a small flat blade screwdriver.

Chemically clean the flow control or the white flow control retainer using dilute sodium bisulfite or vinegar. Do not use a wire brush. If necessary, replace the flow control, o-ring on the flow control retainer, or the o-ring on the elbow.

Reseat the flow control so the rounded end is visible in the flow control. Reseat the white flow control retainer by pushing the retainer into the elbow until the o-ring seats. Remove locking clip, push down on elbow to reseat and insert locking clip.

Do not use Vaseline, oils, or other unacceptable lubricants on o-rings. A silicone lubricant may be used on the o-ring on the elbow or the white retainer.

SPACER STACK ASSEMBLY:

To access the spacer stack assembly remove the drive assembly, drive cap assembly and piston. The spacer stack assembly can be removed easily without tools by using thumb and forefinger. Inspect the black o-rings and clear lip seals for wear or damage. Replace the entire stack if necessary. Do not disassemble the stack.

The spacer stack assembly may be chemically cleaned (dilute sodium bisulfite or vinegar) or wiped with a soft cloth.

The spacer stack assembly can be pushed into the control valve body bore by hand. Use fingers to push the center of the assembly into the control valve body. The assembly is properly seated when at least four threads are exposed (approximately 5/8”). Do not force the spacer stack assembly in. The control valve body bore interior can be lubricated with silicone to allow for easy insertion of the entire stack.

Reattach the drive cap assembly and piston(s) and the drive assembly.

After completing any valve maintenance involving the drive assembly or the drive cap assembly and pistons unplug power source jack from the printed circuit board (black wire) and plug back in or press and hold NEXT and REGEN buttons for 3 seconds.
This resets the electronics and establishes the service piston position. The display should flash all wording, then flash the software version and then reset the valve to the service position.

INJECTOR CAP, SCREEN, INJECTOR PLUG AND INJECTOR:

Unscrew the injector cap and lift off. Loosen cap with special plastic wrench or pliers if necessary. Attached to the injector cap is a screen. Remove the screen and clean if fouled.

The plug and/or injector can be pried out with a small screwdriver. The plug can be wiped clean. If the plug leaks replace the entire plug. The injector consists of a throat and a nozzle. Chemically clean the injector with vinegar or sodium bisulfite. The holes can be blown out with air. Both pieces have small diameter holes that control the flow rates of water to insure that the proper concentration of regenerant is used. Sharp objects, which can score the plastic, should not be used to clean the injector. Scoring the injector or increasing the diameter of the hole could change the operating parameters of the injector.

Push the plug(s) and/or injectors firmly in place, replace the screen and hand tighten the injector cap.

TROUBLESHOOTING

Problem	Possible Cause	Solution
1. No Display on PC Board	a. No power at electric outlet	a. Repair outlet or use working outlet
	b. Control valve Power Adapter not plugged into outlet or power cord end not connected to PC board connection	b. Plug Power Adapter into outlet or connect power cord end to PC Board connection
	c. Improper power supply	c. Verify proper voltage is being delivered to PC Board
	d. Defective Power Adapter	d. Replace Power Adapter
	e. Defective PC Board	e. Replace PC Board
2. PC Board does not display correct time of day	a. Power Adapter plugged into electric outlet controlled by light switch	a. Use uninterrupted outlet
	b. Tripped breaker switch and/or tripped GFI	b. Reset breaker switch and/ or GFI switch
	c. Power outage	c. Reset time of day. If PC Board has battery back up present the battery may be depleted. See Front Cover and Drive Assembly drawing for instructions.
	d. Defective PC Board	d. Replace PC Board
3. Control valve regenerates at wrong time of day	a. Power outage	a. Reset time of day. If PC Board has battery back up present the battery may be depleted. See Front Cover and Drive Assembly drawing for instructions.
	b. Time of day not set correctly	b. Reset to correct time of day
	c. Time of regeneration set incorrectly	c. Reset regeneration time
4. Time of day flashes on and off	a. Power outage	a. Reset time of day. If PC Board has battery back up present the battery may be depleted. See Front Cover and Drive Assembly drawing for instructions.
5. Control valve does not regenerate automatically when the REGEN button is depressed and held	a. Broken drive gear or drive cap assembly	a. Replace drive gear or drive cap assembly
	b. Broken Piston Rod	b. Replace piston rod
	c. Defective PC Board	c. Defective PC Board
6. Control valve does not regenerate automatically but **does** when the REGEN button is depressed and held	a. Bypass valve in bypass position	a. Turn bypass handles to place bypass in service position
	b. Defective PC Board	b. Replace PC Board
7. Untreated water is being delivered	a. Bypass valve is open or faulty	a. Fully close bypass valve or replace
	b. Media is exhausted due to high water usage	b. Check program settings
	c. Water quality fluctuation	c. Test water and adjust program values accordingly
	d. No regenerant or low level of regenerant in regenerant tank	d. Add proper regenerant to tank
	e. Control fails to draw in regenerant	e. Refer to Trouble Shooting Guide number 11
	f. Insufficient regenerant level in regenerant tank	f. Check refill setting in programming. Check refill flow control for restrictions or debris and clean or replace
	g. Damaged seal/stack assembly	g. Replace seal/stack assembly
	h. Control valve body type and piston type mix matched	h. Verify proper control valve body type and piston type match
	i. Fouled media bed	i. Replace media bed
8. Control valve uses too much regenerant	a. Improper refill setting	a. Check refill setting
	b. Improper program settings	b. Check program setting to make sure they are specific to the water quality and application needs
	c. Control valve regenerates frequently	c. Check for leaking fixtures that may be exhausting capacity or system is undersized

Problem	Possible Cause	Solution
9. Residual regenerant being delivered to service	a. Low water pressure	a. Check incoming water pressure – water pressure must remain at minimum of 25 psi
	b. Incorrect injector size	b. Replace injector with correct size for the application
	c. Restricted drain line	c. Check drain line for restrictions or debris and clean
10. Excessive water in regenerant tank	a. Improper program settings	a. Check refill setting
	b. Plugged injector	b. Remove injector and clean or replace
	c. Drive cap assembly not tightened in properly	c. Re-tighten the drive cap assembly
	d. Damaged seal/ stack assembly	d. Replace seal/ stack
	e. Restricted or kinked drain line	e. Check drain line for restrictions or debris and or un-kink drain line
	f. Plugged backwash flow controller	f. Remove backwash flow controller and clean or replace
	g. Missing refill flow controller	g. Replace refill flow controller
11. Control valve fails to draw in regenerant	a. Injector is plugged	a. Remove injector and clean or replace
	b. Faulty regenerant piston	b. Replace regenerant piston
	c. Regenerant line connection leak	c. Inspect regenerant line for air leak
	d. Drain line restriction or debris cause excess back pressure	d. Inspect drain line and clean to correct restriction
	e. Drain line too long or too high	e. Shorten length and or height
	f. Low water pressure	f. Check incoming water pressure – water pressure must remain at minimum of 25 psi
12. Water running to drain	a. Power outage during regeneration	a. Upon power being restored control will finish the remaining regeneration time. Reset time of day.
	b. Damaged seal/ stack assembly	b. Replace seal/ stack assembly
	c. Piston assembly failure	c. Replace piston assembly
	d. Drive cap assembly not tightened in properly	d. Re-tighten the drive cap assembly
13. Err – 101 = Control unable to sense motor movement	a. Motor not inserted full to engage pinion, motor wires broken or disconnected	a. Disconnect power, make sure motor is fully engaged, check for broken wires, make sure two pin connector on motor is connected to the two pin connection on the PC Board labeled MOTOR. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	b. PC Board not properly snapped into drive bracket	b. Properly snap PC Board into drive bracket and then Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	c. Missing reduction gears	c. Replace missing gears

Problem	Possible Cause	Solution
14. Err – 102 = Control valve motor ran too short and was unable to find the next cycle position and stalled	a. Foreign material is lodged in control valve	a. Open up control valve and pull out piston assembly and seal/ stack assembly for inspection. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	b. Mechanical binding	b. Check piston and seal/ stack assembly, check reduction gears, check drive bracket and main drive gear interface. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	c. Main drive gear too tight	c. Loosen main drive gear. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	d. Improper voltage being delivered to PC Board	d. Verify that proper voltage is being supplied. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
15. Err – 103 = Control valve motor ran too long and was unable to find the next cycle position	a. Motor failure during a regeneration	a. Check motor connections then Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	b. Foreign matter built up on piston and stack assemblies creating friction and drag enough to time out motor	b. Replace piston and stack assemblies. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	c. Drive bracket not snapped in properly and out enough that reduction gears and drive gear do not interface	c. Snap drive bracket in properly then Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
16. Err – 104 = Control valve motor ran too long and timed out trying to reach home position	a. Drive bracket not snapped in properly and out enough that reduction gears and drive gear do not interface	a. Snap drive bracket in properly then Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.

Problem	Possible Cause	Solution
17. Err – 106, Err - 116 = MAV/ SEPS/ NHBP/ AUX MAV valve motor ran too long and unable to find the proper park position Motorized Alternating Valve = MAV Separate Source = SEPS No Nitrate Water Bypass = NHBP Auxiliary MAV = AUX MAV	a. Control valve programmed for ALT A or b, nHbP, SEPS, or AUX MAV with out having a MAV or NHBP valve attached to operate that function	a. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect. Then re-program valve to proper setting
	b. MAV/ NHBP motor wire not connected to PC Board	b. Connect MAV/ NHBP motor to PC Board two pin connection labeled DRIVE. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	c. MAV/ NHBP motor not fully engaged with reduction gears	c. Properly insert motor into casing, do not force into casing Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	d. Foreign matter built up on piston and stack assemblies creating friction and drag enough to time out motor	d. Replace piston and stack assemblies. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
18. Err – 107, Err - 117 = MAV/ SEPS/ NHBP/ AUX MAV valve motor ran too short (stalled) while looking for proper park position Motorized Alternating Valve = MAV Separate Source = SEPS No Nitrate Water Bypass = NHBP Auxiliary MAV = AUX MAV	a. Foreign material is lodged in MAV/ NHBP valve	a. Open up MAV/ NHBP valve and check piston and seal/ stack assembly for foreign material. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.
	b. Mechanical binding	b. Check piston and seal/ stack assembly, check reduction gears, drive gear interface, and check MAV/ NHBP black drive pinion on motor for being jammed into motor body. Press NEXT and REGEN buttons for 3 seconds to resynchronize software with piston position or disconnect power supply from PC Board for 5 seconds and then reconnect.



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